CHAPTER 24

# Electric Potential

## 24-1 ELECTRIC POTENTIAL

### Learning Objectives

*After reading this module, you should be able to . . .*

**24.01** Identify that the electric force is conservative and thus has an associated potential energy.

**24.02** Identify that at every point in a charged object's electric field, the object sets up an electric potential $V$, which is a scalar quantity that can be positive or negative depending on the sign of the object's charge.

**24.03** For a charged particle placed at a point in an object's electric field, apply the relationship between the object's electric potential $V$ at that point, the particle's charge $q$, and the potential energy $U$ of the particle–object system.

**24.04** Convert energies between units of joules and electron-volts.

**24.05** If a charged particle moves from an initial point to a final point in an electric field, apply the relationships between the change $\Delta V$ in the potential, the particle's charge $q$, the change $\Delta U$ in the potential energy, and the work $W$ done by the electric force.

**24.06** If a charged particle moves between two given points in the electric field of a charged object, identify that the amount of work done by the electric force is path independent.

**24.07** If a charged particle moves through a change $\Delta V$ in electric potential without an applied force acting on it, relate $\Delta V$ and the change $\Delta K$ in the particle's kinetic energy.

**24.08** If a charged particle moves through a change $\Delta V$ in electric potential while an applied force acts on it, relate $\Delta V$, the change $\Delta K$ in the particle's kinetic energy, and the work $W_{app}$ done by the applied force.

### Key Ideas

● The electric potential $V$ at a point $P$ in the electric field of a charged object is

$$V = \frac{-W_\infty}{q_0} = \frac{U}{q_0},$$

where $W_\infty$ is the work that would be done by the electric force on a positive test charge $q_0$ were it brought from an infinite distance to $P$, and $U$ is the electric potential energy that would then be stored in the test charge–object system.

● If a particle with charge $q$ is placed at a point where the electric potential of a charged object is $V$, the electric potential energy $U$ of the particle–object system is

$$U = qV.$$

● If the particle moves through a potential difference $\Delta V$, the change in the electric potential energy is

$$\Delta U = q\,\Delta V = q(V_f - V_i).$$

● If a particle moves through a change $\Delta V$ in electric potential without an applied force acting on it, applying the conservation of mechanical energy gives the change in kinetic energy as

$$\Delta K = -q\,\Delta V.$$

● If, instead, an applied force acts on the particle, doing work $W_{app}$, the change in kinetic energy is

$$\Delta K = -q\,\Delta V + W_{app}.$$

● In the special case when $\Delta K = 0$, the work of an applied force involves only the motion of the particle through a potential difference:

$$W_{app} = q\,\Delta V.$$

## What Is Physics?

One goal of physics is to identify basic forces in our world, such as the electric force we discussed in Chapter 21. A related goal is to determine whether a force is conservative—that is, whether a potential energy can be associated with it. The motivation for associating a potential energy with a force is that we can then




**Figure 24-1** Particle 1 is located at point $P$ in the electric field of particle 2.

apply the principle of the conservation of mechanical energy to closed systems involving the force. This extremely powerful principle allows us to calculate the results of experiments for which force calculations alone would be very difficult. Experimentally, physicists and engineers discovered that the electric force is conservative and thus has an associated electric potential energy. In this chapter we first define this type of potential energy and then put it to use.

For a quick taste, let's return to the situation we considered in Chapter 22: In Figure 24-1, particle 1 with positive charge $q_1$ is located at point $P$ near particle 2 with positive charge $q_2$. In Chapter 22 we explained how particle 2 is able to push on particle 1 without any contact. To account for the force $\vec{F}$ (which is a vector quantity), we defined an electric field $\vec{E}$ (also a vector quantity) that is set up at $P$ by particle 2. That field exists regardless of whether particle 1 is at $P$. If we choose to place particle 1 there, the push on it is due to charge $q_1$ and that pre-existing field $\vec{E}$.

Here is a related problem. If we release particle 1 at $P$, it begins to move and thus has kinetic energy. Energy cannot appear by magic, so from where does it come? It comes from the electric potential energy $U$ associated with the force between the two particles in the arrangement of Fig. 24-1. To account for the potential energy $U$ (which is a scalar quantity), we define an **electric potential** $V$ (also a scalar quantity) that is set up at $P$ by particle 2. The electric potential exists regardless of whether particle 1 is at $P$. If we choose to place particle 1 there, the potential energy of the two-particle system is then due to charge $q_1$ and that pre-existing electric potential $V$.

Our goals in this chapter are to (1) define electric potential, (2) discuss how to calculate it for various arrangements of charged particles and objects, and (3) discuss how electric potential $V$ is related to electric potential energy $U$.

## Electric Potential and Electric Potential Energy

We are going to define the electric potential (or *potential* for short) in terms of electric potential energy, so our first job is to figure out how to measure that potential energy. Back in Chapter 8, we measured gravitational potential energy $U$ of an object by (1) assigning $U = 0$ for a reference configuration (such as the object at table level) and (2) then calculating the work $W$ the gravitational force does if the object is moved up or down from that level. We then defined the potential energy as being

$$U = -W \quad \text{(potential energy)}. \tag{24-1}$$

Let's follow the same procedure with our new conservative force, the electric force. In Fig. 24-2*a*, we want to find the potential energy $U$ associated with a positive test charge $q_0$ located at point $P$ in the electric field of a charged rod. First, we need a reference configuration for which $U = 0$. A reasonable choice is for the test charge to be infinitely far from the rod, because then there is no interaction with the rod. Next, we bring the test charge in from infinity to point $P$ to form the configuration of Fig. 24-2*a*. Along the way, we calculate the work done by the electric force on the test charge. The potential energy of the final configuration is then given by Eq. 24-1, where $W$ is now the work done by the electric force. Let's use the notation $W_\infty$ to emphasize that the test charge is brought in from infinity. The work and thus the potential energy can be positive or negative depending on the sign of the rod's charge.

Next, we define the electric potential $V$ at $P$ in terms of the work done by the electric force and the resulting potential energy:

$$V = \frac{-W_\infty}{q_0} = \frac{U}{q_0} \quad \text{(electric potential)}. \tag{24-2}$$

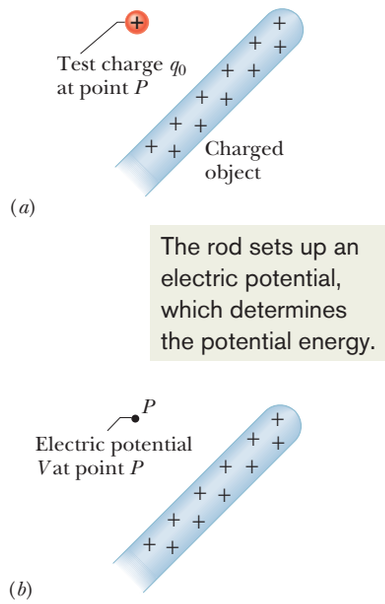



**Figure 24-2** (*a*) A test charge has been brought in from infinity to point $P$ in the electric field of the rod. (*b*) We define an electric potential $V$ at $P$ based on the potential energy of the configuration in (*a*).

That is, the electric potential is the amount of electric potential energy per unit charge when a positive test charge is brought in from infinity. The rod sets up this potential $V$ at $P$ regardless of whether the test charge (or anything else) happens to be there (Fig. 24-2*b*). From Eq. 24-2 we see that $V$ is a scalar quantity (because there is no direction associated with potential energy or charge) and can be positive or negative (because potential energy and charge have signs).

Repeating this procedure we find that an electric potential is set up at every point in the rod's electric field. In fact, every charged object sets up electric potential $V$ at points throughout its electric field. If we happen to place a particle with, say, charge $q$ at a point where we know the pre-existing $V$, we can immediately find the potential energy of the configuration:

$$(\text{electric potential energy}) = (\text{particle's charge})\left(\frac{\text{electric potential energy}}{\text{unit charge}}\right),$$

or
$$U = qV, \tag{24-3}$$

where $q$ can be positive or negative.

***Two Cautions.*** (1) The (now very old) decision to call $V$ a *potential* was unfortunate because the term is easily confused with *potential energy*. Yes, the two quantities are related (that is the point here) but they are very different and not interchangeable. (2) Electric potential is a scalar, not a vector. (When you come to the homework problems, you will rejoice on this point.)

***Language.*** A potential energy is a property of a system (or configuration) of objects, but sometimes we can get away with assigning it to a single object. For example, the gravitational potential energy of a baseball hit to outfield is actually a potential energy of the baseball–Earth system (because it is associated with the force between the baseball and Earth). However, because only the baseball noticeably moves (its motion does not noticeably affect Earth), we might assign the gravitational potential energy to it alone. In a similar way, if a charged particle is placed in an electric field and has no noticeable effect on the field (or the charged object that sets up the field), we usually assign the electric potential energy to the particle alone.

***Units.*** The SI unit for potential that follows from Eq. 24-2 is the joule per coulomb. This combination occurs so often that a special unit, the *volt* (abbreviated V), is used to represent it. Thus,

$$1 \text{ volt} = 1 \text{ joule per coulomb}.$$

With two unit conversions, we can now switch the unit for electric field from newtons per coulomb to a more conventional unit:

$$1 \text{ N/C} = \left(1\frac{\text{N}}{\text{C}}\right)\left(\frac{1 \text{ V}}{1 \text{ J/C}}\right)\left(\frac{1 \text{ J}}{1 \text{ N}\cdot\text{m}}\right)$$
$$= 1 \text{ V/m}.$$

The conversion factor in the second set of parentheses comes from our definition of volt given above; that in the third set of parentheses is derived from the definition of the joule. From now on, we shall express values of the electric field in volts per meter rather than in newtons per coulomb.

## Motion Through an Electric Field

***Change in Electric Potential.*** If we move from an initial point $i$ to a second point $f$ in the electric field of a charged object, the electric potential changes by

$$\Delta V = V_f - V_i.$$

If we move a particle with charge $q$ from $i$ to $f$, then, from Eq. 24-3, the potential energy of the system changes by

$$\Delta U = q\,\Delta V = q(V_f - V_i). \tag{24-4}$$

The change can be positive or negative, depending on the signs of $q$ and $\Delta V$. It can also be zero, if there is no change in potential from $i$ to $f$ (the points have the same value of potential). Because the electric force is conservative, the change in potential energy $\Delta U$ between $i$ and $f$ is the same for all paths between those points (it is *path independent*).

***Work by the Field.*** We can relate the potential energy change $\Delta U$ to the work $W$ done by the electric force as the particle moves from $i$ to $f$ by applying the general relation for a conservative force (Eq. 8-1):

$$W = -\Delta U \quad \text{(work, conservative force)}. \tag{24-5}$$

Next, we can relate that work to the change in the potential by substituting from Eq. 24-4:

$$W = -\Delta U = -q\,\Delta V = -q(V_f - V_i). \tag{24-6}$$

Up until now, we have always attributed work to a force but here can also say that $W$ is the work done on the particle by the electric field (because it, of course, produces the force). The work can be positive, negative, or zero. Because $\Delta U$ between any two points is path independent, so is the work $W$ done by the field. (If you need to calculate work for a difficult path, switch to an easier path—you get the same result.)

***Conservation of Energy.*** If a charged particle moves through an electric field with no force acting on it other than the electric force due to the field, then the mechanical energy is conserved. Let's assume that we can assign the electric potential energy to the particle alone. Then we can write the conservation of mechanical energy of the particle that moves from point $i$ to point $f$ as

$$U_i + K_i = U_f + K_f, \tag{24-7}$$

or

$$\Delta K = -\Delta U. \tag{24-8}$$

Substituting Eq. 24-4, we find a very useful equation for the change in the particle's kinetic energy as a result of the particle moving through a potential difference:

$$\Delta K = -q\,\Delta V = -q(V_f - V_i). \tag{24-9}$$

***Work by an Applied Force.*** If some force in addition to the electric force acts on the particle, we say that the additional force is an *applied force* or *external force*, which is often attributed to an *external agent*. Such an applied force can do work on the particle, but the force may not be conservative and thus, in general, we cannot associate a potential energy with it. We account for that work $W_{\text{app}}$ by modifying Eq. 24-7:

$$\text{(initial energy)} + \text{(work by applied force)} = \text{(final energy)}$$

or

$$U_i + K_i + W_{\text{app}} = U_f + K_f. \tag{24-10}$$

Rearranging and substituting from Eq. 24-4, we can also write this as

$$\Delta K = -\Delta U + W_{\text{app}} = -q\,\Delta V + W_{\text{app}}. \tag{24-11}$$

The work by the applied force can be positive, negative, or zero, and thus the energy of the system can increase, decrease, or remain the same.

In the special case where the particle is stationary before and after the move, the kinetic energy terms in Eqs. 24-10 and 24-11 are zero and we have

$$W_{\text{app}} = q\,\Delta V \quad (\text{for } K_i = K_f). \tag{24-12}$$

In this special case, the work $W_{\text{app}}$ involves the motion of the particle through the potential difference $\Delta V$ and not a change in the particle's kinetic energy.

By comparing Eqs. 24-6 and 24-12, we see that in this special case, the work by the applied force is the negative of the work by the field:

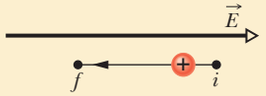
$$W_{app} = -W \quad (\text{for } K_i = K_f). \tag{24-13}$$

***Electron-volts.*** In atomic and subatomic physics, energy measures in the SI unit of joules often require awkward powers of ten. A more convenient (but non-SI unit) is the *electron-volt* (eV), which is defined to be equal to the work required to move a single elementary charge $e$ (such as that of an electron or proton) through a potential difference $\Delta V$ of exactly one volt. From Eq. 24-6, we see that the magnitude of this work is $q\,\Delta V$. Thus,

$$\begin{aligned} 1\text{ eV} &= e(1\text{ V}) \\ &= (1.602 \times 10^{-19}\text{ C})(1\text{ J/C}) = 1.602 \times 10^{-19}\text{ J}. \end{aligned} \tag{24-14}$$

### Checkpoint 1

In the figure, we move a proton from point $i$ to point $f$ in a uniform electric field. Is positive or negative work done by (a) the electric field and (b) our force? (c) Does the electric potential energy increase or decrease? (d) Does the proton move to a point of higher or lower electric potential?




## Sample Problem 24.01 Work and potential energy in an electric field

Electrons are continually being knocked out of air molecules in the atmosphere by cosmic-ray particles coming in from space. Once released, each electron experiences an electric force $\vec{F}$ due to the electric field $\vec{E}$ that is produced in the atmosphere by charged particles already on Earth. Near Earth's surface the electric field has the magnitude $E = 150$ N/C and is directed downward. What is the change $\Delta U$ in the electric potential energy of a released electron when the electric force causes it to move vertically upward through a distance $d = 520$ m (Fig. 24-3)? Through what potential change does the electron move?

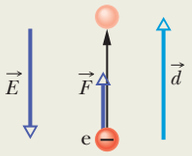



**Figure 24-3** An electron in the atmosphere is moved upward through displacement $\vec{d}$ by an electric force $\vec{F}$ due to an electric field $\vec{E}$.

### KEY IDEAS

(1) The change $\Delta U$ in the electric potential energy of the electron is related to the work $W$ done on the electron by the electric field. Equation 24-5 ($W = -\Delta U$) gives the relation. (2) The work done by a constant force $\vec{F}$ on a particle undergoing a displacement $\vec{d}$ is

$$W = \vec{F} \cdot \vec{d}.$$

(3) The electric force and the electric field are related by the force equation $\vec{F} = q\vec{E}$, where here $q$ is the charge of an electron ($= -1.6 \times 10^{-19}$ C).

***Calculations:*** Substituting the force equation into the work equation and taking the dot product yield

$$W = q\vec{E} \cdot \vec{d} = qEd \cos\theta,$$

where $\theta$ is the angle between the directions of $\vec{E}$ and $\vec{d}$. The field $\vec{E}$ is directed downward and the displacement $\vec{d}$ is directed upward; so $\theta = 180°$. We can now evaluate the work as

$$\begin{aligned} W &= (-1.6 \times 10^{-19}\text{ C})(150\text{ N/C})(520\text{ m}) \cos 180° \\ &= 1.2 \times 10^{-14}\text{ J}. \end{aligned}$$

Equation 24-5 then yields

$$\Delta U = -W = -1.2 \times 10^{-14}\text{ J}. \quad \text{(Answer)}$$

This result tells us that during the 520 m ascent, the electric potential energy of the electron *decreases* by $1.2 \times 10^{-14}$ J. To find the change in electric potential, we apply Eq. 24-4:

$$\begin{aligned} \Delta V = \frac{\Delta U}{-q} &= \frac{-1.2 \times 10^{-14}\text{ J}}{-1.6 \times 10^{-19}\text{ C}} \\ &= 4.5 \times 10^{4}\text{ V} = 45\text{ kV}. \end{aligned} \quad \text{(Answer)}$$

This tells us that the electric force does work to move the electron to a *higher* potential.

# 24-2 EQUIPOTENTIAL SURFACES AND THE ELECTRIC FIELD

## Learning Objectives

*After reading this module, you should be able to . . .*

**24.09** Identify an equipotential surface and describe how it is related to the direction of the associated electric field.

**24.10** Given an electric field as a function of position, calculate the change in potential $\Delta V$ from an initial point to a final point by choosing a path between the points and integrating the dot product of the field $\vec{E}$ and a length element $d\vec{s}$ along the path.

**24.11** For a uniform electric field, relate the field magnitude $E$ and the separation $\Delta x$ and potential difference $\Delta V$ between adjacent equipotential lines.

**24.12** Given a graph of electric field $E$ versus position along an axis, calculate the change in potential $\Delta V$ from an initial point to a final point by graphical integration.

**24.13** Explain the use of a zero-potential location.

## Key Ideas

● The points on an equipotential surface all have the same electric potential. The work done on a test charge in moving it from one such surface to another is independent of the locations of the initial and final points on these surfaces and of the path that joins the points. The electric field $\vec{E}$ is always directed perpendicularly to corresponding equipotential surfaces.

● The electric potential difference between two points $i$ and $f$ is

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s},$$

where the integral is taken over any path connecting the points. If the integration is difficult along any particular path, we can choose a different path along which the integration might be easier.

● If we choose $V_i = 0$, we have, for the potential at a particular point,

$$V = -\int_i^f \vec{E} \cdot d\vec{s}.$$

● In a uniform field of magnitude $E$, the change in potential from a higher equipotential surface to a lower one, separated by distance $\Delta x$, is

$$\Delta V = -E\,\Delta x.$$

## Equipotential Surfaces

Adjacent points that have the same electric potential form an **equipotential surface,** which can be either an imaginary surface or a real, physical surface. No net work $W$ is done on a charged particle by an electric field when the particle moves between two points $i$ and $f$ on the same equipotential surface. This follows from Eq. 24-6, which tells us that $W$ must be zero if $V_f = V_i$. Because of the path independence of work (and thus of potential energy and potential), $W = 0$ for *any* path connecting points $i$ and $f$ on a given equipotential surface regardless of whether that path lies entirely on that surface.

Figure 24-4 shows a *family* of equipotential surfaces associated with the electric field due to some distribution of charges. The work done by the electric field on a charged particle as the particle moves from one end to the other of paths

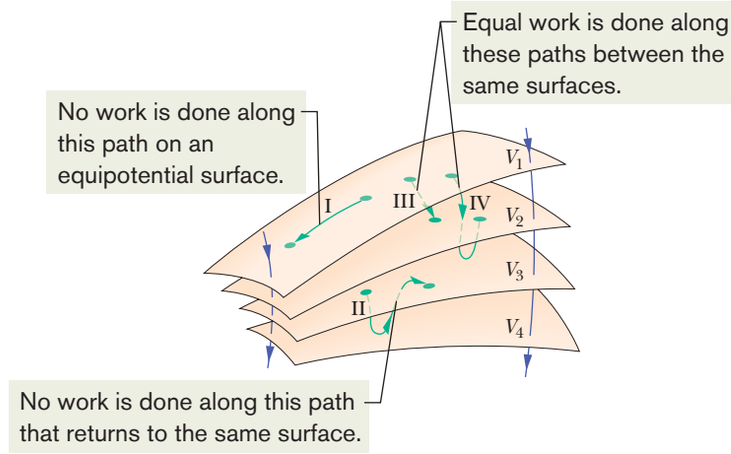



**Figure 24-4** Portions of four equipotential surfaces at electric potentials $V_1 = 100$ V, $V_2 = 80$ V, $V_3 = 60$ V, and $V_4 = 40$ V. Four paths along which a test charge may move are shown. Two electric field lines are also indicated.

I and II is zero because each of these paths begins and ends on the same equipotential surface and thus there is no net change in potential. The work done as the charged particle moves from one end to the other of paths III and IV is not zero but has the same value for both these paths because the initial and final potentials are identical for the two paths; that is, paths III and IV connect the same pair of equipotential surfaces.

From symmetry, the equipotential surfaces produced by a charged particle or a spherically symmetrical charge distribution are a family of concentric spheres. For a uniform electric field, the surfaces are a family of planes perpendicular to the field lines. In fact, equipotential surfaces are always perpendicular to electric field lines and thus to $\vec{E}$, which is always tangent to these lines. If $\vec{E}$ were *not* perpendicular to an equipotential surface, it would have a component lying along that surface. This component would then do work on a charged particle as it moved along the surface. However, by Eq. 24-6 work cannot be done if the surface is truly an equipotential surface; the only possible conclusion is that $\vec{E}$ must be everywhere perpendicular to the surface. Figure 24-5 shows electric field lines and cross sections of the equipotential surfaces for a uniform electric field and for the field associated with a charged particle and with an electric dipole.

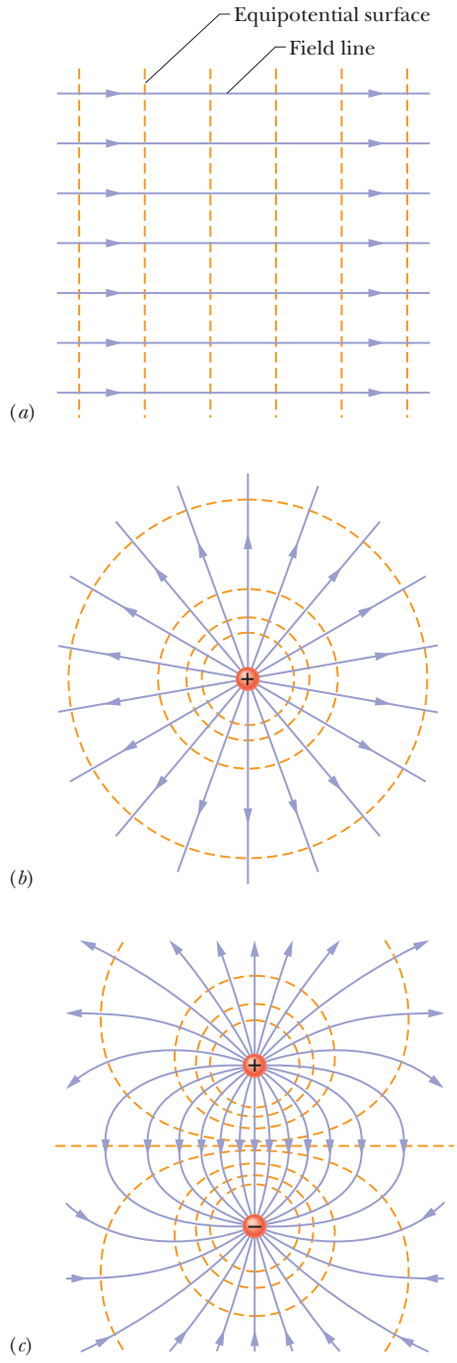



**Figure 24-5** Electric field lines (purple) and cross sections of equipotential surfaces (gold) for (*a*) a uniform electric field, (*b*) the field due to a charged particle, and (*c*) the field due to an electric dipole.

## Calculating the Potential from the Field

We can calculate the potential difference between any two points $i$ and $f$ in an electric field if we know the electric field vector $\vec{E}$ all along any path connecting those points. To make the calculation, we find the work done on a positive test charge by the field as the charge moves from $i$ to $f$, and then use Eq. 24-6.

Consider an arbitrary electric field, represented by the field lines in Fig. 24-6, and a positive test charge $q_0$ that moves along the path shown from point $i$ to point $f$. At any point on the path, an electric force $q_0\vec{E}$ acts on the charge as it moves through a differential displacement $d\vec{s}$. From Chapter 7, we know that the differential work $dW$ done on a particle by a force $\vec{F}$ during a displacement $d\vec{s}$ is given by the dot product of the force and the displacement:

$$dW = \vec{F} \cdot d\vec{s}. \tag{24-15}$$

For the situation of Fig. 24-6, $\vec{F} = q_0\vec{E}$ and Eq. 24-15 becomes

$$dW = q_0\vec{E} \cdot d\vec{s}. \tag{24-16}$$

To find the total work $W$ done on the particle by the field as the particle moves from point $i$ to point $f$, we sum—via integration—the differential works done on the charge as it moves through all the displacements $d\vec{s}$ along the path:

$$W = q_0 \int_i^f \vec{E} \cdot d\vec{s}. \tag{24-17}$$

If we substitute the total work $W$ from Eq. 24-17 into Eq. 24-6, we find

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}. \tag{24-18}$$

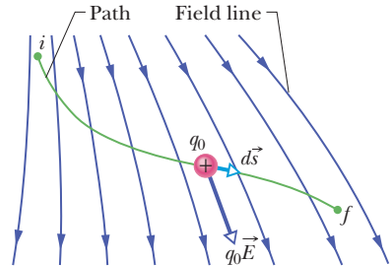



**Figure 24-6** A test charge $q_0$ moves from point $i$ to point $f$ along the path shown in a nonuniform electric field. During a displacement $d\vec{s}$, an electric force $q_0\vec{E}$ acts on the test charge. This force points in the direction of the field line at the location of the test charge.

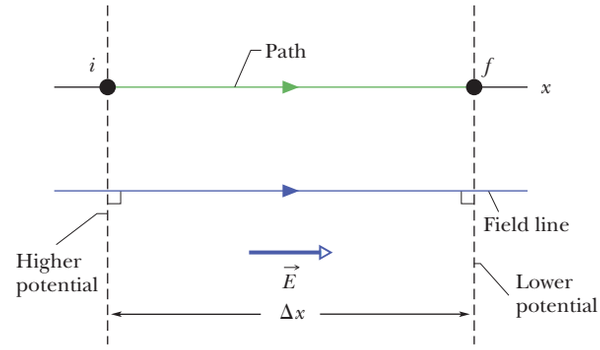



**Figure 24-7** We move between points *i* and *f*, between adjacent equipotential lines in a uniform electric field $\vec{E}$, parallel to a field line.

Thus, the potential difference $V_f - V_i$ between any two points *i* and *f* in an electric field is equal to the negative of the *line integral* (meaning the integral along a particular path) of $\vec{E} \cdot d\vec{s}$ from *i* to *f*. However, because the electric force is conservative, all paths (whether easy or difficult to use) yield the same result.

Equation 24-18 allows us to calculate the difference in potential between any two points in the field. If we set potential $V_i = 0$, then Eq. 24-18 becomes

$$V = -\int_i^f \vec{E} \cdot d\vec{s}, \tag{24-19}$$

in which we have dropped the subscript *f* on $V_f$. Equation 24-19 gives us the potential *V* at any point *f* in the electric field *relative to the zero potential* at point *i*. If we let point *i* be at infinity, then Eq. 24-19 gives us the potential *V* at any point *f* relative to the zero potential at infinity.

***Uniform Field.*** Let's apply Eq. 24-18 for a uniform field as shown in Fig. 24-7. We start at point *i* on an equipotential line with potential $V_i$ and move to point *f* on an equipotential line with a lower potential $V_f$. The separation between the two equipotential lines is $\Delta x$. Let's also move along a path that is parallel to the electric field $\vec{E}$ (and thus perpendicular to the equipotential lines). The angle between $\vec{E}$ and $d\vec{s}$ in Eq. 24-18 is zero, and the dot product gives us

$$\vec{E} \cdot d\vec{s} = E\, ds \cos 0 = E\, ds.$$

Because *E* is constant for a uniform field, Eq. 24-18 becomes

$$V_f - V_i = -E \int_i^f ds. \tag{24-20}$$

The integral is simply an instruction for us to add all the displacement elements *ds* from *i* to *f*, but we already know that the sum is length $\Delta x$. Thus we can write the change in potential $V_f - V_i$ in this uniform field as

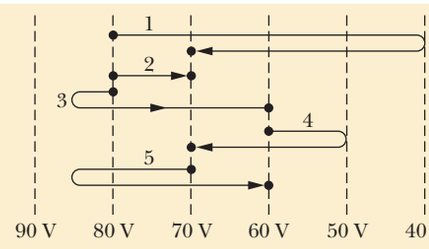
$$\Delta V = -E\, \Delta x \quad \text{(uniform field)}. \tag{24-21}$$

This is the change in voltage $\Delta V$ between two equipotential lines in a uniform field of magnitude *E*, separated by distance $\Delta x$. If we move in the direction of the field by distance $\Delta x$, the potential decreases. In the opposite direction, it increases.

The electric field vector points from higher potential toward lower potential.

## Checkpoint 2

The figure here shows a family of parallel equipotential surfaces (in cross section) and five paths along which we shall move an electron from one surface to another. (a) What is the direction of the electric field associated with the surfaces? (b) For each path, is the work we do positive, negative, or zero? (c) Rank the paths according to the work we do, greatest first.



### Sample Problem 24.02 Finding the potential change from the electric field

(a) Figure 24-8*a* shows two points $i$ and $f$ in a uniform electric field $\vec{E}$. The points lie on the same electric field line (not shown) and are separated by a distance $d$. Find the potential difference $V_f - V_i$ by moving a positive test charge $q_0$ from $i$ to $f$ along the path shown, which is parallel to the field direction.

**KEY IDEA**

We can find the potential difference between any two points in an electric field by integrating $\vec{E} \cdot d\vec{s}$ along a path connecting those two points according to Eq. 24-18.

***Calculations:*** We have actually already done the calculation for such a path in the direction of an electric field line in a uniform field when we derived Eq. 24-21. With slight changes in notation, Eq. 24-21 gives us

$$V_f - V_i = -Ed. \qquad \text{(Answer)}$$

(b) Now find the potential difference $V_f - V_i$ by moving the positive test charge $q_0$ from $i$ to $f$ along the path *icf* shown in Fig. 24-8*b*.

***Calculations:*** The Key Idea of (a) applies here too, except now we move the test charge along a path that consists of two lines: *ic* and *cf*. At all points along line *ic*, the displacement $d\vec{s}$ of the test charge is perpendicular to $\vec{E}$. Thus, the angle $\theta$ between $\vec{E}$ and $d\vec{s}$ is 90°, and the dot product $\vec{E} \cdot d\vec{s}$ is 0. Equation 24-18 then tells us that points $i$ and $c$ are at the same potential: $V_c - V_i = 0$. Ah, we should have seen this coming. The points are on the same equipotential surface, which is perpendicular to the electric field lines.

For line *cf* we have $\theta = 45°$ and, from Eq. 24-18,

$$V_f - V_i = -\int_c^f \vec{E} \cdot d\vec{s} = -\int_c^f E(\cos 45°)\, ds$$

$$= -E(\cos 45°) \int_c^f ds.$$

The integral in this equation is just the length of line *cf*; from Fig. 24-8*b*, that length is $d/\cos 45°$. Thus,

$$V_f - V_i = -E(\cos 45°)\frac{d}{\cos 45°} = -Ed. \qquad \text{(Answer)}$$

This is the same result we obtained in (a), as it must be; the potential difference between two points does not depend on the path connecting them. Moral: When you want to find the potential difference between two points by moving a test charge between them, you can save time and work by choosing a path that simplifies the use of Eq. 24-18.

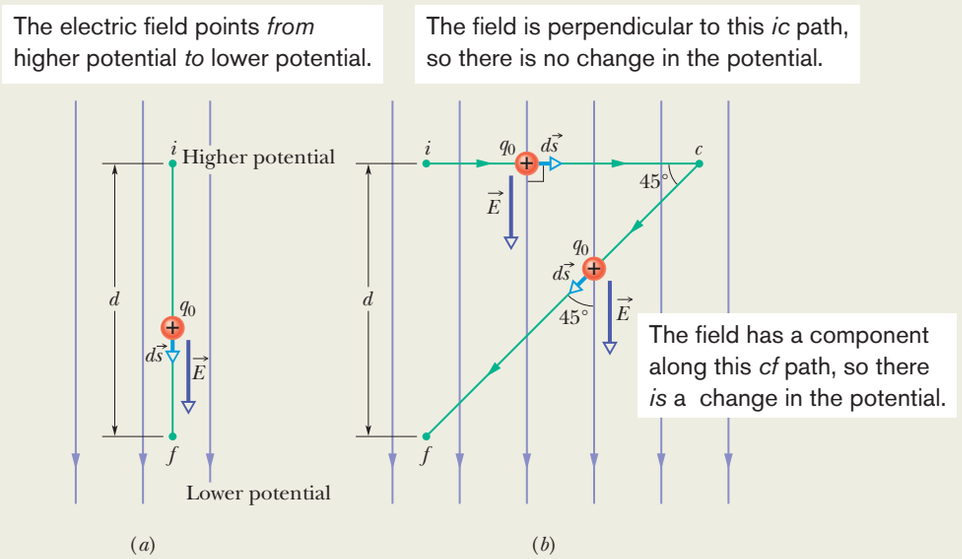



**Figure 24-8** (*a*) A test charge $q_0$ moves in a straight line from point $i$ to point $f$, along the direction of a uniform external electric field. (*b*) Charge $q_0$ moves along path *icf* in the same electric field.

# 24-3 POTENTIAL DUE TO A CHARGED PARTICLE

## Learning Objectives

*After reading this module, you should be able to . . .*

**24.14** For a given point in the electric field of a charged particle, apply the relationship between the electric potential $V$, the charge of the particle $q$, and the distance $r$ from the particle.

**24.15** Identify the correlation between the algebraic signs of the potential set up by a particle and the charge of the particle.

**24.16** For points outside or on the surface of a spherically symmetric charge distribution, calculate the electric potential as if all the charge is concentrated as a particle at the center of the sphere.

**24.17** Calculate the net potential at any given point due to several charged particles, identifying that algebraic addition is used, not vector addition.

**24.18** Draw equipotential lines for a charged particle.

## Key Ideas

● The electric potential due to a single charged particle at a distance $r$ from that charged particle is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r},$$

where $V$ has the same sign as $q$.

● The potential due to a collection of charged particles is

$$V = \sum_{i=1}^{n} V_i = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i}.$$

Thus, the potential is the algebraic sum of the individual potentials, with no consideration of directions.

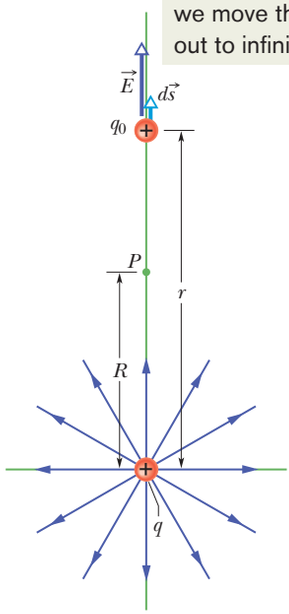



**Figure 24-9** The particle with positive charge $q$ produces an electric field $\vec{E}$ and an electric potential $V$ at point $P$. We find the potential by moving a test charge $q_0$ from $P$ to infinity. The test charge is shown at distance $r$ from the particle, during differential displacement $d\vec{s}$.

## Potential Due to a Charged Particle

We now use Eq. 24-18 to derive, for the space around a charged particle, an expression for the electric potential $V$ relative to the zero potential at infinity. Consider a point $P$ at distance $R$ from a fixed particle of positive charge $q$ (Fig. 24-9). To use Eq. 24-18, we imagine that we move a positive test charge $q_0$ from point $P$ to infinity. Because the path we take does not matter, let us choose the simplest one—a line that extends radially from the fixed particle through $P$ to infinity.

To use Eq. 24-18, we must evaluate the dot product

$$\vec{E}\cdot d\vec{s} = E\cos\theta\, ds. \tag{24-22}$$

The electric field $\vec{E}$ in Fig. 24-9 is directed radially outward from the fixed particle. Thus, the differential displacement $d\vec{s}$ of the test particle along its path has the same direction as $\vec{E}$. That means that in Eq. 24-22, angle $\theta = 0$ and $\cos\theta = 1$. Because the path is radial, let us write $ds$ as $dr$. Then, substituting the limits $R$ and $\infty$, we can write Eq. 24-18 as

$$V_f - V_i = -\int_R^\infty E\, dr. \tag{24-23}$$

Next, we set $V_f = 0$ (at $\infty$) and $V_i = V$ (at $R$). Then, for the magnitude of the electric field at the site of the test charge, we substitute from Eq. 22-3:

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}. \tag{24-24}$$

With these changes, Eq. 24-23 then gives us

$$0 - V = -\frac{q}{4\pi\varepsilon_0}\int_R^\infty \frac{1}{r^2}\,dr = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{r}\right]_R^\infty$$

$$= -\frac{1}{4\pi\varepsilon_0}\frac{q}{R}. \tag{24-25}$$

Solving for $V$ and switching $R$ to $r$, we then have

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \tag{24-26}$$

as the electric potential $V$ due to a particle of charge $q$ at any radial distance $r$ from the particle.

Although we have derived Eq. 24-26 for a positively charged particle, the derivation holds also for a negatively charged particle, in which case, $q$ is a negative quantity. Note that the sign of $V$ is the same as the sign of $q$:

A positively charged particle produces a positive electric potential. A negatively charged particle produces a negative electric potential.

Figure 24-10 shows a computer-generated plot of Eq. 24-26 for a positively charged particle; the magnitude of $V$ is plotted vertically. Note that the magnitude increases as $r \to 0$. In fact, according to Eq. 24-26, $V$ is infinite at $r = 0$, although Fig. 24-10 shows a finite, smoothed-off value there.

Equation 24-26 also gives the electric potential either *outside or on the external surface of* a spherically symmetric charge distribution. We can prove this by using one of the shell theorems of Modules 21-1 and 23-6 to replace the actual spherical charge distribution with an equal charge concentrated at its center. Then the derivation leading to Eq. 24-26 follows, provided we do not consider a point within the actual distribution.

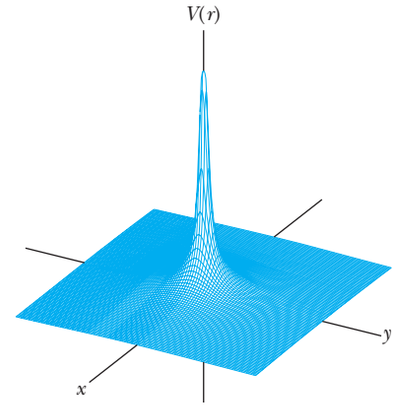



**Figure 24-10** A computer-generated plot of the electric potential $V(r)$ due to a positively charged particle located at the origin of an $xy$ plane. The potentials at points in the $xy$ plane are plotted vertically. (Curved lines have been added to help you visualize the plot.) The infinite value of $V$ predicted by Eq. 24-26 for $r = 0$ is not plotted.

## Potential Due to a Group of Charged Particles

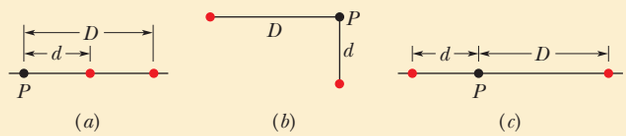
We can find the net electric potential at a point due to a group of charged particles with the help of the superposition principle. Using Eq. 24-26 with the plus or minus sign of the charge included, we calculate separately the potential resulting from each charge at the given point. Then we sum the potentials. Thus, for $n$ charges, the net potential is

$$V = \sum_{i=1}^{n} V_i = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i} \quad (n \text{ charged particles}). \tag{24-27}$$

Here $q_i$ is the value of the $i$th charge and $r_i$ is the radial distance of the given point from the $i$th charge. The sum in Eq. 24-27 is an *algebraic sum,* not a vector sum like the sum that would be used to calculate the electric field resulting from a group of charged particles. Herein lies an important computational advantage of potential over electric field: It is a lot easier to sum several scalar quantities than to sum several vector quantities whose directions and components must be considered.

### Checkpoint 3

The figure here shows three arrangements of two protons. Rank the arrangements according to the net electric potential produced at point $P$ by the protons, greatest first.



### Sample Problem 24.03 Net potential of several charged particles

What is the electric potential at point $P$, located at the center of the square of charged particles shown in Fig. 24-11*a*? The distance $d$ is 1.3 m, and the charges are

$$q_1 = +12\text{ nC}, \quad q_3 = +31\text{ nC},$$
$$q_2 = -24\text{ nC}, \quad q_4 = +17\text{ nC}.$$

#### KEY IDEA

The electric potential $V$ at point $P$ is the algebraic sum of the electric potentials contributed by the four particles.

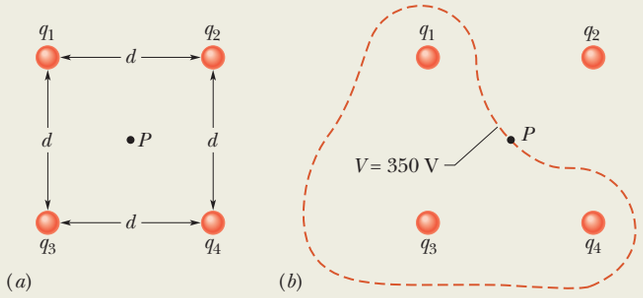



**Figure 24-11** (*a*) Four charged particles. (*b*) The closed curve is a (roughly drawn) cross section of the equipotential surface that contains point $P$.

(Because electric potential is a scalar, the orientations of the particles do not matter.)

***Calculations:*** From Eq. 24-27, we have

$$V = \sum_{i=1}^{4} V_i = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r} + \frac{q_2}{r} + \frac{q_3}{r} + \frac{q_4}{r}\right).$$

The distance $r$ is $d/\sqrt{2}$, which is 0.919 m, and the sum of the charges is

$$q_1 + q_2 + q_3 + q_4 = (12 - 24 + 31 + 17) \times 10^{-9}\text{ C}$$
$$= 36 \times 10^{-9}\text{ C}.$$

Thus,
$$V = \frac{(8.99 \times 10^9\text{ N}\cdot\text{m}^2/\text{C}^2)(36 \times 10^{-9}\text{ C})}{0.919\text{ m}}$$
$$\approx 350\text{ V}. \qquad \text{(Answer)}$$

Close to any of the three positively charged particles in Fig. 24-11*a*, the potential has very large positive values. Close to the single negative charge, the potential has very large negative values. Therefore, there must be points within the square that have the same intermediate potential as that at point $P$. The curve in Fig. 24-11*b* shows the intersection of the plane of the figure with the equipotential surface that contains point $P$.

### Sample Problem 24.04 Potential is not a vector, orientation is irrelevant

(a) In Fig. 24-12*a*, 12 electrons (of charge $-e$) are equally spaced and fixed around a circle of radius $R$. Relative to $V = 0$ at infinity, what are the electric potential and electric field at the center $C$ of the circle due to these electrons?

#### KEY IDEAS

(1) The electric potential $V$ at $C$ is the algebraic sum of the electric potentials contributed by all the electrons. Because electric potential is a scalar, the orientations of the electrons do not matter. (2) The electric field at $C$ is a vector quantity and thus the orientation of the electrons *is* important.

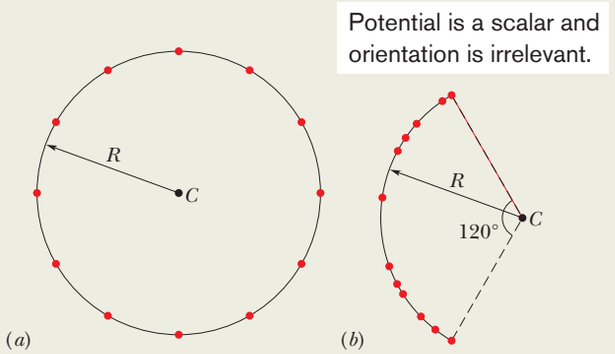



**Figure 24-12** (*a*) Twelve electrons uniformly spaced around a circle. (*b*) The electrons nonuniformly spaced along an arc of the original circle.

***Calculations:*** Because the electrons all have the same negative charge $-e$ and are all the same distance $R$ from $C$, Eq. 24-27 gives us

$$V = -12\,\frac{1}{4\pi\varepsilon_0}\,\frac{e}{R}. \qquad \text{(Answer)} \quad (24\text{-}28)$$

Because of the symmetry of the arrangement in Fig. 24-12*a*, the electric field vector at $C$ due to any given electron is canceled by the field vector due to the electron that is diametrically opposite it. Thus, at $C$,

$$\vec{E} = 0. \qquad \text{(Answer)}$$

(b) The electrons are moved along the circle until they are nonuniformly spaced over a 120° arc (Fig. 24-12*b*). At $C$, find the electric potential and describe the electric field.

***Reasoning:*** The potential is still given by Eq. 24-28, because the distance between $C$ and each electron is unchanged and orientation is irrelevant. The electric field is no longer zero, however, because the arrangement is no longer symmetric. A net field is now directed toward the charge distribution.

# 24-4 POTENTIAL DUE TO AN ELECTRIC DIPOLE

## Learning Objectives

*After reading this module, you should be able to . . .*

**24.19** Calculate the potential $V$ at any given point due to an electric dipole, in terms of the magnitude $p$ of the dipole moment or the product of the charge separation $d$ and the magnitude $q$ of either charge.

**24.20** For an electric dipole, identify the locations of positive potential, negative potential, and zero potential.

**24.21** Compare the decrease in potential with increasing distance for a single charged particle and an electric dipole.

## Key Idea

● At a distance $r$ from an electric dipole with dipole moment magnitude $p = qd$, the electric potential of the dipole is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}$$

for $r \gg d$; the angle $\theta$ lies between the dipole moment vector and a line extending from the dipole midpoint to the point of measurement.

## Potential Due to an Electric Dipole

Now let us apply Eq. 24-27 to an electric dipole to find the potential at an arbitrary point $P$ in Fig. 24-13*a*. At $P$, the positively charged particle (at distance $r_{(+)}$) sets up potential $V_{(+)}$ and the negatively charged particle (at distance $r_{(-)}$) sets up potential $V_{(-)}$. Then the net potential at $P$ is given by Eq. 24-27 as

$$V = \sum_{i=1}^{2} V_i = V_{(+)} + V_{(-)} = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_{(+)}} + \frac{-q}{r_{(-)}}\right)$$

$$= \frac{q}{4\pi\varepsilon_0}\frac{r_{(-)} - r_{(+)}}{r_{(-)}r_{(+)}}. \qquad (24\text{-}29)$$

Naturally occurring dipoles—such as those possessed by many molecules—are quite small; so we are usually interested only in points that are relatively far from the dipole, such that $r \gg d$, where $d$ is the distance between the charges and $r$ is the distance from the dipole's midpoint to $P$. In that case, we can approximate the two lines to $P$ as being parallel and their length difference as being the leg of a right triangle with hypotenuse $d$ (Fig. 24-13*b*). Also, that difference is so small that the product of the lengths is approximately $r^2$. Thus,

$$r_{(-)} - r_{(+)} \approx d\cos\theta \quad \text{and} \quad r_{(-)}r_{(+)} \approx r^2.$$

If we substitute these quantities into Eq. 24-29, we can approximate $V$ to be

$$V = \frac{q}{4\pi\varepsilon_0}\frac{d\cos\theta}{r^2},$$

where $\theta$ is measured from the dipole axis as shown in Fig. 24-13*a*. We can now write $V$ as

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2} \quad \text{(electric dipole)}, \qquad (24\text{-}30)$$

in which $p$ $(= qd)$ is the magnitude of the electric dipole moment $\vec{p}$ defined in Module 22-3. The vector $\vec{p}$ is directed along the dipole axis, from the negative to the positive charge. (Thus, $\theta$ is measured from the direction of $\vec{p}$.) We use this vector to report the orientation of an electric dipole.

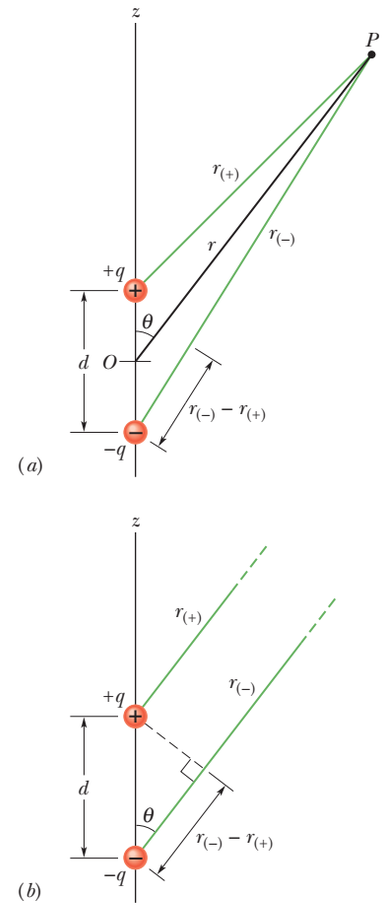
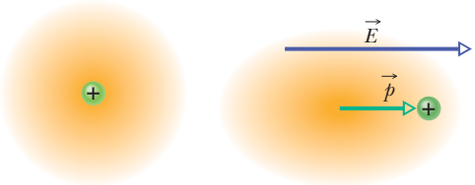



**Figure 24-13** (*a*) Point $P$ is a distance $r$ from the midpoint $O$ of a dipole. The line $OP$ makes an angle $\theta$ with the dipole axis. (*b*) If $P$ is far from the dipole, the lines of lengths $r_{(+)}$ and $r_{(-)}$ are approximately parallel to the line of length $r$, and the dashed black line is approximately perpendicular to the line of length $r_{(-)}$.

The electric field shifts the positive and negative charges, creating a dipole.




(*a*) (*b*)

**Figure 24-14** (*a*) An atom, showing the positively charged nucleus (green) and the negatively charged electrons (gold shading). The centers of positive and negative charge coincide. (*b*) If the atom is placed in an external electric field $\vec{E}$, the electron orbits are distorted so that the centers of positive and negative charge no longer coincide. An induced dipole moment $\vec{p}$ appears. The distortion is greatly exaggerated here.

### Checkpoint 4

Suppose that three points are set at equal (large) distances $r$ from the center of the dipole in Fig. 24-13: Point $a$ is on the dipole axis above the positive charge, point $b$ is on the axis below the negative charge, and point $c$ is on a perpendicular bisector through the line connecting the two charges. Rank the points according to the electric potential of the dipole there, greatest (most positive) first.

### Induced Dipole Moment

Many molecules, such as water, have *permanent* electric dipole moments. In other molecules (called *nonpolar molecules*) and in every isolated atom, the centers of the positive and negative charges coincide (Fig. 24-14*a*) and thus no dipole moment is set up. However, if we place an atom or a nonpolar molecule in an external electric field, the field distorts the electron orbits and separates the centers of positive and negative charge (Fig. 24-14*b*). Because the electrons are negatively charged, they tend to be shifted in a direction opposite the field. This shift sets up a dipole moment $\vec{p}$ that points in the direction of the field. This dipole moment is said to be *induced* by the field, and the atom or molecule is then said to be *polarized* by the field (that is, it has a positive side and a negative side). When the field is removed, the induced dipole moment and the polarization disappear.

# 24-5 POTENTIAL DUE TO A CONTINUOUS CHARGE DISTRIBUTION

## Learning Objective

*After reading this module, you should be able to . . .*

**24.22** For charge that is distributed uniformly along a line or over a surface, find the net potential at a given point by splitting the distribution up into charge elements and summing (by integration) the potential due to each one.

## Key Ideas

- For a continuous distribution of charge (over an extended object), the potential is found by (1) dividing the distribution into charge elements $dq$ that can be treated as particles and then (2) summing the potential due to each element by integrating over the full distribution:

$$V = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r}.$$

- In order to carry out the integration, $dq$ is replaced with the product of either a linear charge density $\lambda$ and a length element (such as $dx$), or a surface charge density $\sigma$ and area element (such as $dx\,dy$).
- In some cases where the charge is symmetrically distributed, a two-dimensional integration can be reduced to a one-dimensional integration.

## Potential Due to a Continuous Charge Distribution

When a charge distribution $q$ is continuous (as on a uniformly charged thin rod or disk), we cannot use the summation of Eq. 24-27 to find the potential $V$ at a point $P$. Instead, we must choose a differential element of charge $dq$, determine the potential $dV$ at $P$ due to $dq$, and then integrate over the entire charge distribution.

Let us again take the zero of potential to be at infinity. If we treat the element of charge $dq$ as a particle, then we can use Eq. 24-26 to express the potential $dV$ at point $P$ due to $dq$:

$$dV = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r} \quad \text{(positive or negative } dq\text{)}. \tag{24-31}$$

Here $r$ is the distance between $P$ and $dq$. To find the total potential $V$ at $P$, we

integrate to sum the potentials due to all the charge elements:

$$V = \int dV = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r}. \tag{24-32}$$

The integral must be taken over the entire charge distribution. Note that because the electric potential is a scalar, there are *no vector components* to consider in Eq. 24-32.

We now examine two continuous charge distributions, a line and a disk.

## Line of Charge

In Fig. 24-15*a*, a thin nonconducting rod of length $L$ has a positive charge of uniform linear density $\lambda$. Let us determine the electric potential $V$ due to the rod at point $P$, a perpendicular distance $d$ from the left end of the rod.

We consider a differential element $dx$ of the rod as shown in Fig. 24-15*b*. This (or any other) element of the rod has a differential charge of

$$dq = \lambda\, dx. \tag{24-33}$$

This element produces an electric potential $dV$ at point $P$, which is a distance $r = (x^2 + d^2)^{1/2}$ from the element (Fig. 24-15*c*). Treating the element as a point charge, we can use Eq. 24-31 to write the potential $dV$ as

$$dV = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, dx}{(x^2 + d^2)^{1/2}}. \tag{24-34}$$

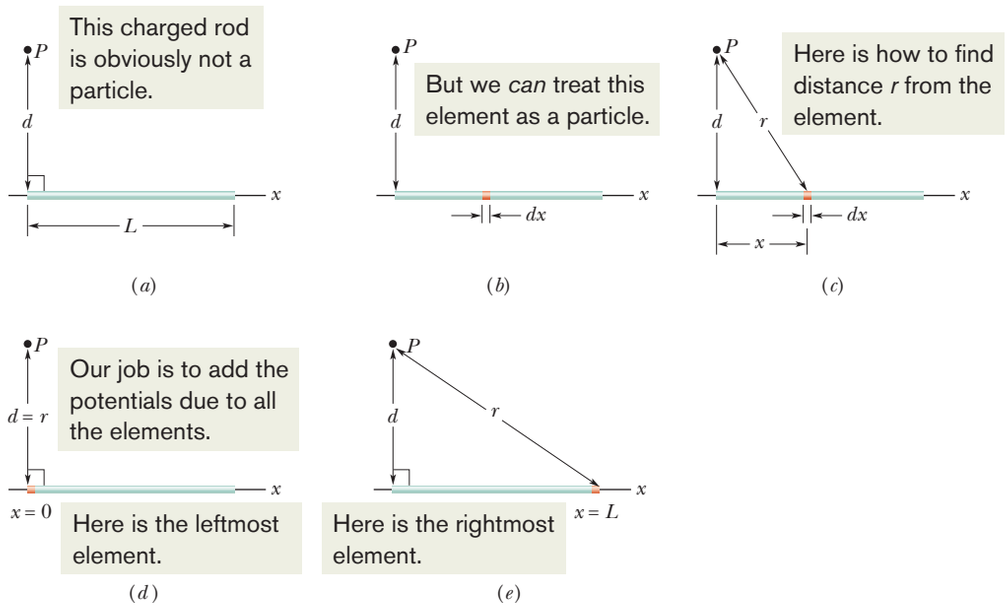



**Figure 24-15** (*a*) A thin, uniformly charged rod produces an electric potential $V$ at point $P$. (*b*) An element can be treated as a particle. (*c*) The potential at $P$ due to the element depends on the distance $r$. We need to sum the potentials due to all the elements, from the left side (*d*) to the right side (*e*).

Since the charge on the rod is positive and we have taken $V = 0$ at infinity, we know from Module 24-3 that $dV$ in Eq. 24-34 must be positive.

We now find the total potential $V$ produced by the rod at point $P$ by integrating Eq. 24-34 along the length of the rod, from $x = 0$ to $x = L$ (Figs. 24-15*d* and *e*), using integral 17 in Appendix E. We find

$$V = \int dV = \int_0^L \frac{1}{4\pi\varepsilon_0}\frac{\lambda}{(x^2 + d^2)^{1/2}}\,dx$$

$$= \frac{\lambda}{4\pi\varepsilon_0}\int_0^L \frac{dx}{(x^2 + d^2)^{1/2}}$$

$$= \frac{\lambda}{4\pi\varepsilon_0}\left[\ln\left(x + (x^2 + d^2)^{1/2}\right)\right]_0^L$$

$$= \frac{\lambda}{4\pi\varepsilon_0}\left[\ln\left(L + (L^2 + d^2)^{1/2}\right) - \ln d\right].$$

We can simplify this result by using the general relation $\ln A - \ln B = \ln(A/B)$. We then find

$$V = \frac{\lambda}{4\pi\varepsilon_0}\ln\left[\frac{L + (L^2 + d^2)^{1/2}}{d}\right]. \tag{24-35}$$

Because $V$ is the sum of positive values of $dV$, it too is positive, consistent with the logarithm being positive for an argument greater than 1.

## Charged Disk

In Module 22-5, we calculated the magnitude of the electric field at points on the central axis of a plastic disk of radius $R$ that has a uniform charge density $\sigma$ on one surface. Here we derive an expression for $V(z)$, the electric potential at any point on the central axis. Because we have a circular distribution of charge on the disk, we could start with a differential element that occupies angle $d\theta$ and radial distance $dr$. We would then need to set up a two-dimensional integration. However, let's do something easier.

In Fig. 24-16, consider a differential element consisting of a flat ring of radius $R'$ and radial width $dR'$. Its charge has magnitude

$$dq = \sigma(2\pi R')(dR'),$$

in which $(2\pi R')(dR')$ is the upper surface area of the ring. All parts of this charged element are the same distance $r$ from point $P$ on the disk's axis. With the aid of Fig. 24-16, we can use Eq. 24-31 to write the contribution of this ring to the electric potential at $P$ as

$$dV = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r} = \frac{1}{4\pi\varepsilon_0}\frac{\sigma(2\pi R')(dR')}{\sqrt{z^2 + R'^2}}. \tag{24-36}$$

We find the net potential at $P$ by adding (via integration) the contributions of all the rings from $R' = 0$ to $R' = R$:

$$V = \int dV = \frac{\sigma}{2\varepsilon_0}\int_0^R \frac{R'\,dR'}{\sqrt{z^2 + R'^2}} = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{z^2 + R^2} - z\right). \tag{24-37}$$

Note that the variable in the second integral of Eq. 24-37 is $R'$ and not $z$, which remains constant while the integration over the surface of the disk is carried out. (Note also that, in evaluating the integral, we have assumed that $z \geq 0$.)

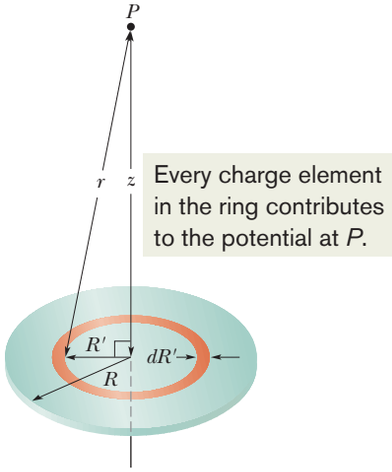



**Figure 24-16** A plastic disk of radius $R$, charged on its top surface to a uniform surface charge density $\sigma$. We wish to find the potential $V$ at point $P$ on the central axis of the disk.

# 24-6 CALCULATING THE FIELD FROM THE POTENTIAL

## Learning Objectives

*After reading this module, you should be able to . . .*

**24.23** Given an electric potential as a function of position along an axis, find the electric field along that axis.
**24.24** Given a graph of electric potential versus position along an axis, determine the electric field along the axis.
**24.25** For a uniform electric field, relate the field magnitude $E$ and the separation $\Delta x$ and potential difference $\Delta V$ between adjacent equipotential lines.
**24.26** Relate the direction of the electric field and the directions in which the potential decreases and increases.

## Key Ideas

- The component of $\vec{E}$ in any direction is the negative of the rate at which the potential changes with distance in that direction:

$$E_s = -\frac{\partial V}{\partial s}.$$

- The $x$, $y$, and $z$ components of $\vec{E}$ may be found from

$$E_x = -\frac{\partial V}{\partial x}; \qquad E_y = -\frac{\partial V}{\partial y}; \qquad E_z = -\frac{\partial V}{\partial z}.$$

When $\vec{E}$ is uniform, all this reduces to

$$E = -\frac{\Delta V}{\Delta s},$$

where $s$ is perpendicular to the equipotential surfaces.

- The electric field is zero parallel to an equipotential surface.

## Calculating the Field from the Potential

In Module 24-2, you saw how to find the potential at a point $f$ if you know the electric field along a path from a reference point to point $f$. In this module, we propose to go the other way—that is, to find the electric field when we know the potential. As Fig. 24-5 shows, solving this problem graphically is easy: If we know the potential $V$ at all points near an assembly of charges, we can draw in a family of equipotential surfaces. The electric field lines, sketched perpendicular to those surfaces, reveal the variation of $\vec{E}$. What we are seeking here is the mathematical equivalent of this graphical procedure.

Figure 24-17 shows cross sections of a family of closely spaced equipotential surfaces, the potential difference between each pair of adjacent surfaces being $dV$. As the figure suggests, the field $\vec{E}$ at any point $P$ is perpendicular to the equipotential surface through $P$.

Suppose that a positive test charge $q_0$ moves through a displacement $d\vec{s}$ from one equipotential surface to the adjacent surface. From Eq. 24-6, we see that the work the electric field does on the test charge during the move is $-q_0\,dV$. From Eq. 24-16 and Fig. 24-17, we see that the work done by the electric field may also be written as the scalar product $(q_0\vec{E})\cdot d\vec{s}$, or $q_0E(\cos\theta)\,ds$. Equating these two expressions for the work yields

$$-q_0\,dV = q_0E(\cos\theta)\,ds, \tag{24-38}$$

or

$$E\cos\theta = -\frac{dV}{ds}. \tag{24-39}$$

Since $E\cos\theta$ is the component of $\vec{E}$ in the direction of $d\vec{s}$, Eq. 24-39 becomes

$$E_s = -\frac{\partial V}{\partial s}. \tag{24-40}$$

We have added a subscript to $E$ and switched to the partial derivative symbols to emphasize that Eq. 24-40 involves only the variation of $V$ along a specified axis (here called the $s$ axis) and only the component of $\vec{E}$ along that axis. In words, Eq. 24-40 (which is essentially the reverse operation of Eq. 24-18) states:

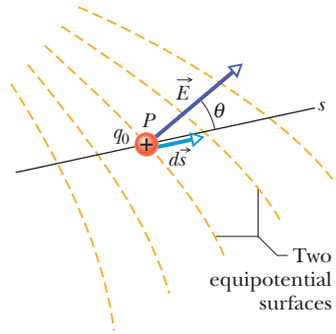



**Figure 24-17** A test charge $q_0$ moves a distance $d\vec{s}$ from one equipotential surface to another. (The separation between the surfaces has been exaggerated for clarity.) The displacement $d\vec{s}$ makes an angle $\theta$ with the direction of the electric field $\vec{E}$.

The component of $\vec{E}$ in any direction is the negative of the rate at which the electric potential changes with distance in that direction.

If we take the $s$ axis to be, in turn, the $x$, $y$, and $z$ axes, we find that the $x$, $y$, and $z$ components of $\vec{E}$ at any point are

$$E_x = -\frac{\partial V}{\partial x}; \qquad E_y = -\frac{\partial V}{\partial y}; \qquad E_z = -\frac{\partial V}{\partial z}. \tag{24-41}$$

Thus, if we know $V$ for all points in the region around a charge distribution—that is, if we know the function $V(x, y, z)$—we can find the components of $\vec{E}$, and thus $\vec{E}$ itself, at any point by taking partial derivatives.

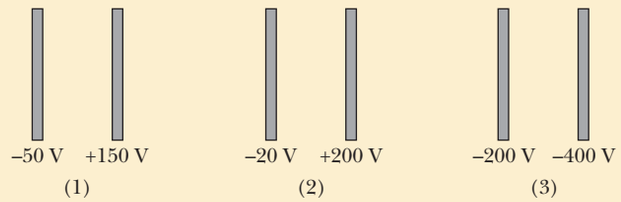
For the simple situation in which the electric field $\vec{E}$ is uniform, Eq. 24-40 becomes

$$E = -\frac{\Delta V}{\Delta s}, \tag{24-42}$$

where $s$ is perpendicular to the equipotential surfaces. The component of the electric field is zero in any direction parallel to the equipotential surfaces because there is no change in potential along the surfaces.

### Checkpoint 5

The figure shows three pairs of parallel plates with the same separation, and the electric potential of each plate. The electric field between the plates is uniform and perpendicular to the plates. (a) Rank the pairs according to the magnitude of the electric field between the plates, greatest first. (b) For which pair is the electric field pointing rightward? (c) If an electron is released midway between the third pair of plates, does it remain there, move rightward at constant speed, move leftward at constant speed, accelerate rightward, or accelerate leftward?




## Sample Problem 24.05 Finding the field from the potential

The electric potential at any point on the central axis of a uniformly charged disk is given by Eq. 24-37,

$$V = \frac{\sigma}{2\varepsilon_0}(\sqrt{z^2 + R^2} - z).$$

Starting with this expression, derive an expression for the electric field at any point on the axis of the disk.

### KEY IDEAS

We want the electric field $\vec{E}$ as a function of distance $z$ along the axis of the disk. For any value of $z$, the direction of $\vec{E}$ must be along that axis because the disk has circular symmetry about that axis. Thus, we want the component $E_z$ of $\vec{E}$ in the direction of $z$. This component is the negative of the rate at which the electric potential changes with distance $z$.

***Calculation:*** Thus, from the last of Eqs. 24-41, we can write

$$E_z = -\frac{\partial V}{\partial z} = -\frac{\sigma}{2\varepsilon_0}\frac{d}{dz}(\sqrt{z^2 + R^2} - z)$$

$$= \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right). \tag{Answer}$$

This is the same expression that we derived in Module 22-5 by integration, using Coulomb's law.

# 24-7 ELECTRIC POTENTIAL ENERGY OF A SYSTEM OF CHARGED PARTICLES

## Learning Objectives

*After reading this module, you should be able to . . .*

**24.27** Identify that the total potential energy of a system of charged particles is equal to the work an applied force must do to assemble the system, starting with the particles infinitely far apart.
**24.28** Calculate the potential energy of a pair of charged particles.
**24.29** Identify that if a system has more than two charged particles, then the system's total potential energy is equal to the sum of the potential energies of every pair of the particles.
**24.30** Apply the principle of the conservation of mechanical energy to a system of charged particles.
**24.31** Calculate the escape speed of a charged particle from a system of charged particles (the minimum initial speed required to move infinitely far from the system).

## Key Idea

● The electric potential energy of a system of charged particles is equal to the work needed to assemble the system with the particles initially at rest and infinitely distant from each other. For two particles at separation $r$,

$$U = W = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}.$$

## Electric Potential Energy of a System of Charged Particles

In this module we are going to calculate the potential energy of a system of two charged particles and then briefly discuss how to expand the result to a system of more than two particles. Our starting point is to examine the work we must do (as an external agent) to bring together two charged particles that are initially infinitely far apart and that end up near each other and stationary. If the two particles have the same sign of charge, we must fight against their mutual repulsion. Our work is then positive and results in a positive potential energy for the final two-particle system. If, instead, the two particles have opposite signs of charge, our job is easy because of the mutual attraction of the particles. Our work is then negative and results in a negative potential energy for the system.

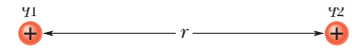



**Figure 24-18** Two charges held a fixed distance $r$ apart.

Let's follow this procedure to build the two-particle system in Fig. 24-18, where particle 1 (with positive charge $q_1$) and particle 2 (with positive charge $q_2$) have separation $r$. Although both particles are positively charged, our result will apply also to situations where they are both negatively charged or have different signs.

We start with particle 2 fixed in place and particle 1 infinitely far away, with an initial potential energy $U_i$ for the two-particle system. Next we bring particle 1 to its final position, and then the system's potential energy is $U_f$. Our work changes the system's potential energy by $\Delta U = U_f - U_i$.

With Eq. 24-4 ($\Delta U = q(V_f - V_i)$), we can relate $\Delta U$ to the change in potential through which we move particle 1:

$$U_f - U_i = q_1(V_f - V_i). \tag{24-43}$$

Let's evaluate these terms. The initial potential energy is $U_i = 0$ because the particles are in the reference configuration (as discussed in Module 24-1). The two potentials in Eq. 24-43 are due to particle 2 and are given by Eq. 24-26:

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r}. \tag{24-44}$$

This tells us that when particle 1 is initially at distance $r = \infty$, the potential at its location is $V_i = 0$. When we move it to the final position at distance $r$, the potential at its location is

$$V_f = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r}. \tag{24-45}$$

Substituting these results into Eq. 24-43 and dropping the subscript $f$, we find that the final configuration has a potential energy of

$$U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r} \quad \text{(two-particle system).} \tag{24-46}$$

Equation 24-46 includes the signs of the two charges. If the two charges have the same sign, $U$ is positive. If they have opposite signs, $U$ is negative.

If we next bring in a third particle, with charge $q_3$, we repeat our calculation, starting with particle 3 at an infinite distance and then bringing it to a final position at distance $r_{31}$ from particle 1 and distance $r_{32}$ from particle 2. At the final position, the potential $V_f$ at the location of particle 3 is the algebraic sum of the potential $V_1$ due to particle 1 and the potential $V_2$ of particle 2. When we work out the algebra, we find that



The total potential energy of a system of particles is the sum of the potential energies for every pair of particles in the system.

This result applies to a system for any given number of particles.

Now that we have an expression for the potential energy of a system of particles, we can apply the principle of the conservation of energy to the system as expressed in Eq. 24-10. For example, if the system consists of many particles, we might consider the kinetic energy (and the associated *escape speed*) required of one of the particles to escape from the rest of the particles.



## Sample Problem 24.06 Potential energy of a system of three charged particles

Figure 24-19 shows three charged particles held in fixed positions by forces that are not shown. What is the electric potential energy $U$ of this system of charges? Assume that $d = 12$ cm and that

$$q_1 = +q, \quad q_2 = -4q, \quad \text{and} \quad q_3 = +2q,$$

in which $q = 150$ nC.

### KEY IDEA

The potential energy $U$ of the system is equal to the work we must do to assemble the system, bringing in each charge from an infinite distance.

***Calculations:*** Let's mentally build the system of Fig. 24-19, starting with one of the charges, say $q_1$, in place and the others at infinity. Then we bring another one, say $q_2$, in from infinity and put it in place. From Eq. 24-46 with $d$ substituted for $r$, the potential energy $U_{12}$ associated with the pair of charges $q_1$ and $q_2$ is

$$U_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{d}.$$

We then bring the last charge $q_3$ in from infinity and put it in place. The work that we must do in this last step is equal to the sum of the work we must do to bring $q_3$ near $q_1$ and the work we must do to bring it near $q_2$. From Eq. 24-46, with $d$ substituted for $r$, that sum is

$$W_{13} + W_{23} = U_{13} + U_{23} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_3}{d} + \frac{1}{4\pi\varepsilon_0}\frac{q_2 q_3}{d}.$$

The total potential energy $U$ of the three-charge system is the sum of the potential energies associated with the three pairs of charges. This sum (which is actually independent of the order in which the charges are brought together) is

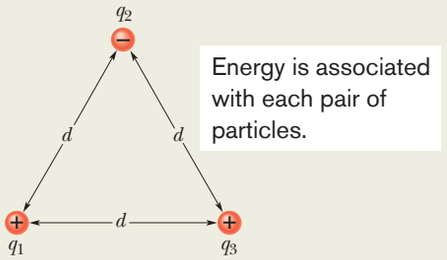



**Figure 24-19** Three charges are fixed at the vertices of an equilateral triangle. What is the electric potential energy of the system?

$$U = U_{12} + U_{13} + U_{23}$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{(+q)(-4q)}{d} + \frac{(+q)(+2q)}{d} + \frac{(-4q)(+2q)}{d}\right)$$

$$= -\frac{10q^2}{4\pi\varepsilon_0 d}$$

$$= -\frac{(8.99 \times 10^9\ \mathrm{N \cdot m^2/C^2})(10)(150 \times 10^{-9}\ \mathrm{C})^2}{0.12\ \mathrm{m}}$$

$$= -1.7 \times 10^{-2}\ \mathrm{J} = -17\ \mathrm{mJ}. \qquad \text{(Answer)}$$

The negative potential energy means that negative work would have to be done to assemble this structure, starting with the three charges infinitely separated and at rest. Put another way, an external agent would have to do 17 mJ of positive work to disassemble the structure completely, ending with the three charges infinitely far apart.

The lesson here is this: If you are given an assembly of charged particles, you can find the potential energy of the assembly by finding the potential of every possible pair of the particles and then summing the results.

## Sample Problem 24.07 Conservation of mechanical energy with electric potential energy

An alpha particle (two protons, two neutrons) moves into a stationary gold atom (79 protons, 118 neutrons), passing through the electron region that surrounds the gold nucleus like a shell and headed directly toward the nucleus (Fig. 24-20). The alpha particle slows until it momentarily stops when its center is at radial distance $r = 9.23$ fm from the nuclear center. Then it moves back along its incoming path. (Because the gold nucleus is much more massive than the alpha particle, we can assume the gold nucleus does not move.) What was the kinetic energy $K_i$ of the alpha particle when it was initially far away (hence external to the gold atom)? Assume that the only force acting between the alpha particle and the gold nucleus is the (electrostatic) Coulomb force and treat each as a single charged particle.

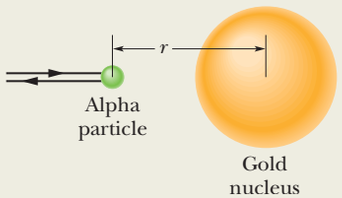



**Figure 24-20** An alpha particle, traveling head-on toward the center of a gold nucleus, comes to a momentary stop (at which time all its kinetic energy has been transferred to electric potential energy) and then reverses its path.

### KEY IDEA

During the entire process, the mechanical energy of the *alpha particle + gold atom* system is conserved.

***Reasoning:*** When the alpha particle is outside the atom, the system's initial electric potential energy $U_i$ is zero because the atom has an equal number of electrons and protons, which produce a *net* electric field of zero. However, once the alpha particle passes through the electron region surrounding the nucleus on its way to the nucleus, the electric field due to the electrons goes to zero. The reason is that the electrons act like a closed spherical shell of uniform negative charge and, as discussed in Module 23-6, such a shell produces zero electric field in the space it encloses. The alpha particle still experiences the electric field of the protons in the nucleus, which produces a repulsive force on the protons within the alpha particle.

As the incoming alpha particle is slowed by this repulsive force, its kinetic energy is transferred to electric potential energy of the system. The transfer is complete when the alpha particle momentarily stops and the kinetic energy is $K_f = 0$.

***Calculations:*** The principle of conservation of mechanical energy tells us that

$$K_i + U_i = K_f + U_f. \qquad (24\text{-}47)$$

We know two values: $U_i = 0$ and $K_f = 0$. We also know that the potential energy $U_f$ at the stopping point is given by the right side of Eq. 24-46, with $q_1 = 2e$, $q_2 = 79e$ (in which $e$ is the elementary charge, $1.60 \times 10^{-19}$ C), and $r = 9.23$ fm. Thus, we can rewrite Eq. 24-47 as

$$K_i = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(79e)}{9.23\ \mathrm{fm}}$$

$$= \frac{(8.99 \times 10^9\ \mathrm{N \cdot m^2/C^2})(158)(1.60 \times 10^{-19}\ \mathrm{C})^2}{9.23 \times 10^{-15}\ \mathrm{m}}$$

$$= 3.94 \times 10^{-12}\ \mathrm{J} = 24.6\ \mathrm{MeV}. \qquad \text{(Answer)}$$

# 24-8 POTENTIAL OF A CHARGED ISOLATED CONDUCTOR

## Learning Objectives

*After reading this module, you should be able to . . .*

**24.32** Identify that an excess charge placed on an isolated conductor (or connected isolated conductors) will distribute itself on the surface of the conductor so that all points of the conductor come to the same potential.

**24.33** For an isolated spherical conducting shell, sketch graphs of the potential and the electric field magnitude versus distance from the center, both inside and outside the shell.

**24.34** For an isolated spherical conducting shell, identify that internally the electric field is zero and the electric potential has the same value as the surface and that externally the electric field and the electric potential have values as though all of the shell's charge is concentrated as a particle at its center.

**24.35** For an isolated cylindrical conducting shell, identify that internally the electric field is zero and the electric potential has the same value as the surface and that externally the electric field and the electric potential have values as though all of the cylinder's charge is concentrated as a line of charge on the central axis.

## Key Ideas

- An excess charge placed on a conductor will, in the equilibrium state, be located entirely on the outer surface of the conductor.
- The entire conductor, including interior points, is at a uniform potential.
- If an isolated charged conductor is placed in an external electric field, then at every internal point, the electric field due to the charge cancels the external electric field that otherwise would have been there.
- Also, the net electric field at every point on the surface is perpendicular to the surface.

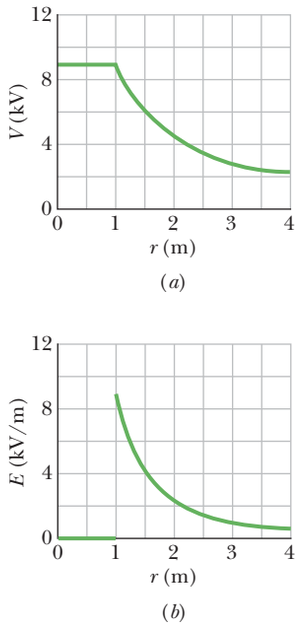



**Figure 24-21** (*a*) A plot of $V(r)$ both inside and outside a charged spherical shell of radius 1.0 m. (*b*) A plot of $E(r)$ for the same shell.

## Potential of a Charged Isolated Conductor

In Module 23-3, we concluded that $\vec{E} = 0$ for all points inside an isolated conductor. We then used Gauss' law to prove that an excess charge placed on an isolated conductor lies entirely on its surface. (This is true even if the conductor has an empty internal cavity.) Here we use the first of these facts to prove an extension of the second:

> An excess charge placed on an isolated conductor will distribute itself on the surface of that conductor so that all points of the conductor—whether on the surface or inside—come to the same potential. This is true even if the conductor has an internal cavity and even if that cavity contains a net charge.

Our proof follows directly from Eq. 24-18, which is

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}.$$

Since $\vec{E} = 0$ for all points within a conductor, it follows directly that $V_f = V_i$ for all possible pairs of points $i$ and $f$ in the conductor.

Figure 24-21*a* is a plot of potential against radial distance $r$ from the center for an isolated spherical conducting shell of 1.0 m radius, having a charge of 1.0 $\mu$C. For points outside the shell, we can calculate $V(r)$ from Eq. 24-26 because the charge $q$ behaves for such external points as if it were concentrated at the center of the shell. That equation holds right up to the surface of the shell. Now let us push a small test charge through the shell—assuming a small hole exists—to its center. No extra work is needed to do this because no net electric force acts on the test charge once it is inside the shell. Thus, the potential at all points inside the shell has the same value as that on the surface, as Fig. 24-21*a* shows.

Figure 24-21*b* shows the variation of electric field with radial distance for the same shell. Note that $E = 0$ everywhere inside the shell. The curves of Fig. 24-21*b* can be derived from the curve of Fig. 24-21*a* by differentiating with respect to $r$, using Eq. 24-40 (recall that the derivative of any constant is zero). The curve of Fig. 24-21*a* can be derived from the curves of Fig. 24-21*b* by integrating with respect to $r$, using Eq. 24-19.

### Spark Discharge from a Charged Conductor

On nonspherical conductors, a surface charge does not distribute itself uniformly over the surface of the conductor. At sharp points or sharp edges, the surface charge density—and thus the external electric field, which is proportional to it—may reach very high values. The air around such sharp points or edges may become ionized, producing the corona discharge that golfers and mountaineers see on the tips of bushes, golf clubs, and rock hammers when thunderstorms threaten. Such corona discharges, like hair that stands on end, are often the precursors of lightning strikes. In such circumstances, it is wise to enclose yourself in a cavity inside a conducting shell, where the electric field is guaranteed to be zero. A car (unless it is a convertible or made with a plastic body) is almost ideal (Fig. 24-22).



Courtesy Westinghouse Electric Corporation

**Figure 24-22** A large spark jumps to a car's body and then exits by moving across the insulating left front tire (note the flash there), leaving the person inside unharmed.

### Isolated Conductor in an External Electric Field

If an isolated conductor is placed in an *external electric field,* as in Fig. 24-23, all points of the conductor still come to a single potential regardless of whether the conductor has an excess charge. The free conduction electrons distribute themselves on the surface in such a way that the electric field they produce at interior points cancels the external electric field that would otherwise be there. Furthermore, the electron distribution causes the net electric field at all points on the surface to be perpendicular to the surface. If the conductor in Fig. 24-23 could be somehow removed, leaving the surface charges frozen in place, the internal and external electric field would remain absolutely unchanged.

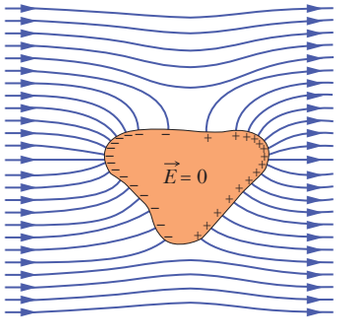



**Figure 24-23** An uncharged conductor is suspended in an external electric field. The free electrons in the conductor distribute themselves on the surface as shown, so as to reduce the net electric field inside the conductor to zero and make the net field at the surface perpendicular to the surface.

## Review & Summary

**Electric Potential** The electric potential $V$ at a point $P$ in the electric field of a charged object is

$$V = \frac{-W_\infty}{q_0} = \frac{U}{q_0}, \tag{24-2}$$

where $W_\infty$ is the work that would be done by the electric force on a positive test charge were it brought from an infinite distance to $P$, and $U$ is the potential energy that would then be stored in the test charge–object system.

**Electric Potential Energy** If a particle with charge $q$ is placed at a point where the electric potential of a charged object is $V$, the electric potential energy $U$ of the particle–object system is

$$U = qV. \tag{24-3}$$

If the particle moves through a potential difference $\Delta V$, the change in the electric potential energy is

$$\Delta U = q\,\Delta V = q(V_f - V_i). \tag{24-4}$$

**Mechanical Energy** If a particle moves through a change $\Delta V$ in electric potential without an applied force acting on it, applying the conservation of mechanical energy gives the change in kinetic energy as

$$\Delta K = -q\,\Delta V. \tag{24-9}$$

If, instead, an applied force acts on the particle, doing work $W_{app}$, the change in kinetic energy is

$$\Delta K = -q\,\Delta V + W_{app}. \tag{24-11}$$

In the special case when $\Delta K = 0$, the work of an applied force

involves only the motion of the particle through a potential difference:

$$W_{app} = q\,\Delta V \quad (\text{for } K_i = K_f). \tag{24-12}$$

**Equipotential Surfaces** The points on an **equipotential surface** all have the same electric potential. The work done on a test charge in moving it from one such surface to another is independent of the locations of the initial and final points on these surfaces and of the path that joins the points. The electric field $\vec{E}$ is always directed perpendicularly to corresponding equipotential surfaces.

**Finding $V$ from $\vec{E}$** The electric potential difference between two points $i$ and $f$ is

$$V_f - V_i = -\int_i^f \vec{E}\cdot d\vec{s}, \tag{24-18}$$

where the integral is taken over any path connecting the points. If the integration is difficult along any particular path, we can choose a different path along which the integration might be easier. If we choose $V_i = 0$, we have, for the potential at a particular point,

$$V = -\int_i^f \vec{E}\cdot d\vec{s}. \tag{24-19}$$

In the special case of a uniform field of magnitude $E$, the potential change between two adjacent (parallel) equipotential lines separated by distance $\Delta x$ is

$$\Delta V = -E\,\Delta x. \tag{24-21}$$

**Potential Due to a Charged Particle** The electric potential due to a single charged particle at a distance $r$ from that particle is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}, \tag{24-26}$$

where $V$ has the same sign as $q$. The potential due to a collection of charged particles is

$$V = \sum_{i=1}^{n} V_i = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i}. \tag{24-27}$$

**Potential Due to an Electric Dipole** At a distance $r$ from an electric dipole with dipole moment magnitude $p = qd$, the electric potential of the dipole is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2} \tag{24-30}$$

for $r \gg d$; the angle $\theta$ is defined in Fig. 24-13.

**Potential Due to a Continuous Charge Distribution** For a continuous distribution of charge, Eq. 24-27 becomes

$$V = \frac{1}{4\pi\varepsilon_0}\int\frac{dq}{r}, \tag{24-32}$$

in which the integral is taken over the entire distribution.

**Calculating $\vec{E}$ from $V$** The component of $\vec{E}$ in any direction is the negative of the rate at which the potential changes with distance in that direction:

$$E_s = -\frac{\partial V}{\partial s}. \tag{24-40}$$

The $x$, $y$, and $z$ components of $\vec{E}$ may be found from

$$E_x = -\frac{\partial V}{\partial x}; \qquad E_y = -\frac{\partial V}{\partial y}; \qquad E_z = -\frac{\partial V}{\partial z}. \tag{24-41}$$

When $\vec{E}$ is uniform, Eq. 24-40 reduces to

$$E = -\frac{\Delta V}{\Delta s}, \tag{24-42}$$

where $s$ is perpendicular to the equipotential surfaces.

**Electric Potential Energy of a System of Charged Particles** The electric potential energy of a system of charged particles is equal to the work needed to assemble the system with the particles initially at rest and infinitely distant from each other. For two particles at separation $r$,

$$U = W = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}. \tag{24-46}$$

**Potential of a Charged Conductor** An excess charge placed on a conductor will, in the equilibrium state, be located entirely on the outer surface of the conductor. The charge will distribute itself so that the following occur: (1) The entire conductor, including interior points, is at a uniform potential. (2) At every internal point, the electric field due to the charge cancels the external electric field that otherwise would have been there. (3) The net electric field at every point on the surface is perpendicular to the surface.

## Questions

**1** Figure 24-24 shows eight particles that form a square, with distance $d$ between adjacent particles. What is the net electric potential at point $P$ at the center of the square if we take the electric potential to be zero at infinity?

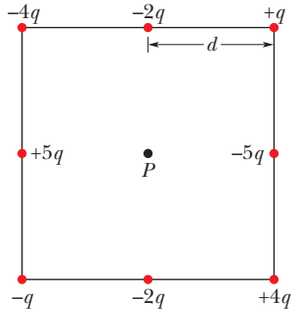



**Figure 24-24** Question 1.

**2** Figure 24-25 shows three sets of cross sections of equipotential surfaces in uniform electric fields; all three cover the same size region of space. The electric potential is indicated for each equipotential surface. (a) Rank the arrangements according to the magnitude of the electric field present in the region, greatest first. (b) In which is the electric field directed down the page?

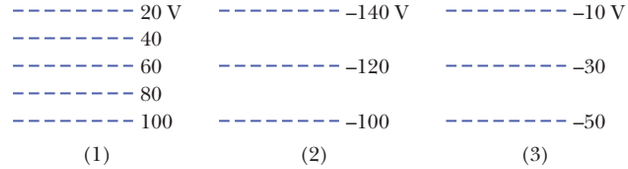



**Figure 24-25** Question 2.

**3** Figure 24-26 shows four pairs of charged particles. For each pair, let $V = 0$ at infinity and consider $V_{net}$ at points on the $x$ axis. For which pairs is there a point at which $V_{net} = 0$ (a) between the particles and (b) to the right of the particles? (c) At such a point is $\vec{E}_{net}$ due to the particles equal to zero? (d) For each pair, are there off-axis points (other than at infinity) where $V_{net} = 0$?

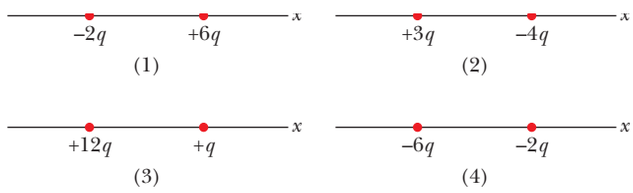



**Figure 24-26** Questions 3 and 9.

**4** Figure 24-27 gives the electric potential $V$ as a function of $x$. (a) Rank the five regions according to the magnitude of the $x$ component of the electric field within them, greatest first. What is the direction of the field along the $x$ axis in (b) region 2 and (c) region 4?

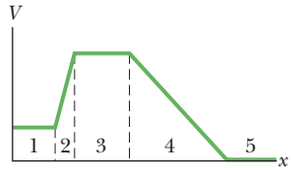



**Figure 24-27** Question 4.

**5** Figure 24-28 shows three paths along which we can move the positively charged sphere $A$ closer to positively charged sphere $B$, which is held fixed in place. (a) Would sphere $A$ be moved to a higher or lower electric potential? Is the work done (b) by our force and (c) by the electric field due to $B$ positive, negative, or zero? (d) Rank the paths according to the work our force does, greatest first.

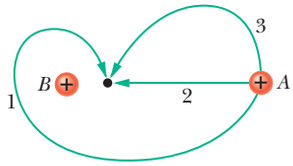



**Figure 24-28** Question 5.

**6** Figure 24-29 shows four arrangements of charged particles, all the same distance from the origin. Rank the situations according to the net electric potential at the origin, most positive first. Take the potential to be zero at infinity.

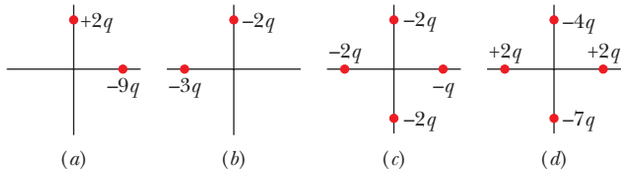



**Figure 24-29** Question 6.

**7** Figure 24-30 shows a system of three charged particles. If you move the particle of charge $+q$ from point $A$ to point $D$, are the following quantities positive, negative, or zero: (a) the change in the electric potential energy of the three-particle system, (b) the work done by the net electric force on the particle you moved (that is, the net force due to the other two particles), and (c) the work done by your force? (d) What are the answers to (a) through (c) if, instead, the particle is moved from $B$ to $C$?

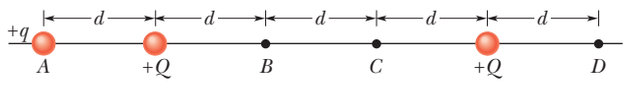
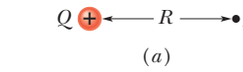



**Figure 24-30** Questions 7 and 8.

**8** In the situation of Question 7, is the work done by your force positive, negative, or zero if the particle is moved (a) from $A$ to $B$, (b) from $A$ to $C$, and (c) from $B$ to $D$? (d) Rank those moves according to the magnitude of the work done by your force, greatest first.

**9** Figure 24-26 shows four pairs of charged particles with identical separations. (a) Rank the pairs according to their electric potential energy (that is, the energy of the two-particle system), greatest (most positive) first. (b) For each pair, if the separation between the particles is increased, does the potential energy of the pair increase or decrease?




**10** (a) In Fig. 24-31$a$, what is the potential at point $P$ due to charge $Q$ at distance $R$ from $P$? Set $V = 0$ at infinity. (b) In Fig. 24-31$b$, the same charge $Q$ has been spread uniformly over a circular arc of radius $R$ and central angle 40°. What is the potential at point $P$, the center of curvature of the arc? (c) In Fig. 24-31$c$, the same charge $Q$ has been spread uniformly over a circle of radius $R$. What is the potential at point $P$, the center of the circle? (d) Rank the three situations according to the magnitude of the electric field that is set up at $P$, greatest first.

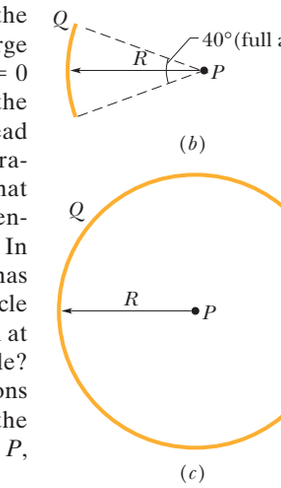



**Figure 24-31** Question 10.

**11** Figure 24-32 shows a thin, uniformly charged rod and three points at the same distance $d$ from the rod. Rank the magnitude of the electric potential the rod produces at those three points, greatest first.

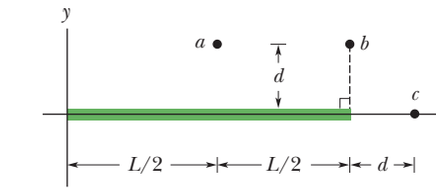



**Figure 24-32** Question 11.

**12** In Fig. 24-33, a particle is to be released at rest at point $A$ and then is to be accelerated directly through point $B$ by an electric field. The potential difference between points $A$ and $B$ is 100 V. Which point should be at higher electric potential if the particle is (a) an electron, (b) a proton, and (c) an alpha particle (a nucleus of two protons and two neutrons)? (d) Rank the kinetic energies of the particles at point $B$, greatest first.

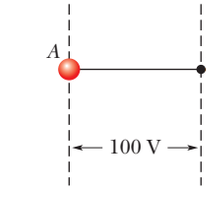



**Figure 24-33**
Question 12.

# Problems

GO Tutoring problem available (at instructor's discretion) in *WileyPLUS* and WebAssign
SSM Worked-out solution available in Student Solutions Manual
• – ••• Number of dots indicates level of problem difficulty
Additional information available in *The Flying Circus of Physics* and at flyingcircusofphysics.com
WWW Worked-out solution is at
ILW Interactive solution is at
http://www.wiley.com/college/halliday

## Module 24-1 Electric Potential

•1 SSM A particular 12 V car battery can send a total charge of 84 A · h (ampere-hours) through a circuit, from one terminal to the other. (a) How many coulombs of charge does this represent? (*Hint:* See Eq. 21-3.) (b) If this entire charge undergoes a change in electric potential of 12 V, how much energy is involved?

•2 The electric potential difference between the ground and a cloud in a particular thunderstorm is $1.2 \times 10^9$ V. In the unit electron-volts, what is the magnitude of the change in the electric potential energy of an electron that moves between the ground and the cloud?

•3 Suppose that in a lightning flash the potential difference between a cloud and the ground is $1.0 \times 10^9$ V and the quantity of charge transferred is 30 C. (a) What is the change in energy of that transferred charge? (b) If all the energy released could be used to accelerate a 1000 kg car from rest, what would be its final speed?

## Module 24-2 Equipotential Surfaces and the Electric Field

•4 Two large, parallel, conducting plates are 12 cm apart and have charges of equal magnitude and opposite sign on their facing surfaces. An electric force of $3.9 \times 10^{-15}$ N acts on an electron placed anywhere between the two plates. (Neglect fringing.) (a) Find the electric field at the position of the electron. (b) What is the potential difference between the plates?

•5 SSM An infinite nonconducting sheet has a surface charge density $\sigma = 0.10\ \mu\text{C/m}^2$ on one side. How far apart are equipotential surfaces whose potentials differ by 50 V?

•6 When an electron moves from $A$ to $B$ along an electric field line in Fig. 24-34, the electric field does $3.94 \times 10^{-19}$ J of work on it. What are the electric potential differences (a) $V_B - V_A$, (b) $V_C - V_A$, and (c) $V_C - V_B$?

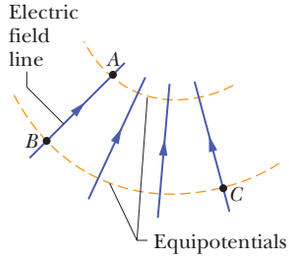


Figure 24-34 Problem 6.

••7 The electric field in a region of space has the components $E_y = E_z = 0$ and $E_x = (4.00\ \text{N/C})x$. Point $A$ is on the $y$ axis at $y = 3.00$ m, and point $B$ is on the $x$ axis at $x = 4.00$ m. What is the potential difference $V_B - V_A$?

••8 A graph of the $x$ component of the electric field as a function of $x$ in a region of space is shown in Fig. 24-35. The scale of the vertical axis is set by $E_{xs} = 20.0$ N/C. The $y$ and $z$ components of the electric field are zero in this region. If the electric potential at the origin is 10 V, (a) what is the electric potential at $x = 2.0$ m, (b) what is the greatest positive value of the electric potential for points on the $x$ axis for which $0 \le x \le 6.0$ m, and (c) for what value of $x$ is the electric potential zero?

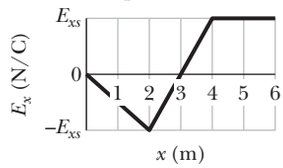


Figure 24-35 Problem 8.

••9 An infinite nonconducting sheet has a surface charge density $\sigma = +5.80\ \text{pC/m}^2$. (a) How much work is done by the electric field due to the sheet if a particle of charge $q = +1.60 \times 10^{-19}$ C is moved from the sheet to a point $P$ at distance $d = 3.56$ cm from the sheet? (b) If the electric potential $V$ is defined to be zero on the sheet, what is $V$ at $P$?

•••10 GO Two uniformly charged, infinite, nonconducting planes are parallel to a $yz$ plane and positioned at $x = -50$ cm and $x = +50$ cm. The charge densities on the planes are $-50\ \text{nC/m}^2$ and $+25\ \text{nC/m}^2$, respectively. What is the magnitude of the potential difference between the origin and the point on the $x$ axis at $x = +80$ cm? (*Hint:* Use Gauss' law.)

•••11 A nonconducting sphere has radius $R = 2.31$ cm and uniformly distributed charge $q = +3.50$ fC. Take the electric potential at the sphere's center to be $V_0 = 0$. What is $V$ at radial distance (a) $r = 1.45$ cm and (b) $r = R$. (*Hint:* See Module 23-6.)

## Module 24-3 Potential Due to a Charged Particle

•12 As a space shuttle moves through the dilute ionized gas of Earth's ionosphere, the shuttle's potential is typically changed by −1.0 V during one revolution. Assuming the shuttle is a sphere of radius 10 m, estimate the amount of charge it collects.

•13 What are (a) the charge and (b) the charge density on the surface of a conducting sphere of radius 0.15 m whose potential is 200 V (with $V = 0$ at infinity)?

•14 Consider a particle with charge $q = 1.0\ \mu$C, point $A$ at distance $d_1 = 2.0$ m from $q$, and point $B$ at distance $d_2 = 1.0$ m. (a) If $A$ and $B$ are diametrically opposite each other, as in Fig. 24-36*a*, what is the electric potential difference $V_A - V_B$? (b) What is that electric potential difference if $A$ and $B$ are located as in Fig. 24-36*b*?

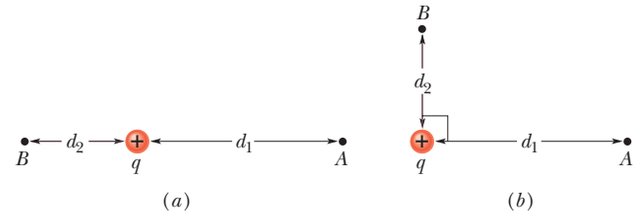


Figure 24-36 Problem 14.

••15 SSM ILW A spherical drop of water carrying a charge of 30 pC has a potential of 500 V at its surface (with $V = 0$ at infinity). (a) What is the radius of the drop? (b) If two such drops of the same charge and radius combine to form a single spherical drop, what is the potential at the surface of the new drop?

••16 GO Figure 24-37 shows a rectangular array of charged particles fixed in place, with distance $a = 39.0$ cm and the charges shown as integer multiples of $q_1 = 3.40$ pC and $q_2 = 6.00$ pC. With $V = 0$ at infinity, what

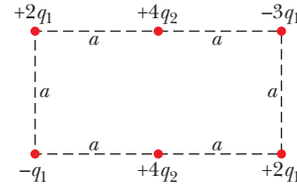


Figure 24-37 Problem 16.

is the net electric potential at the rectangle's center? (*Hint:* Thoughtful examination of the arrangement can reduce the calculation.)

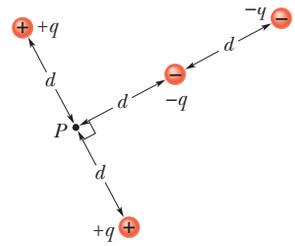



**Figure 24-38** Problem 17.

**••17** GO In Fig. 24-38, what is the net electric potential at point $P$ due to the four particles if $V = 0$ at infinity, $q = 5.00$ fC, and $d = 4.00$ cm?

**••18** GO Two charged particles are shown in Fig. 24-39*a*. Particle 1, with charge $q_1$, is fixed in place at distance $d$. Particle 2, with charge $q_2$, can be moved along the $x$ axis. Figure 24-39*b* gives the net electric potential $V$ at the origin due to the two particles as a function of the $x$ coordinate of particle 2. The scale of the $x$ axis is set by $x_s = 16.0$ cm. The plot has an asymptote of $V = 5.76 \times 10^{-7}$ V as $x \to \infty$. What is $q_2$ in terms of $e$?

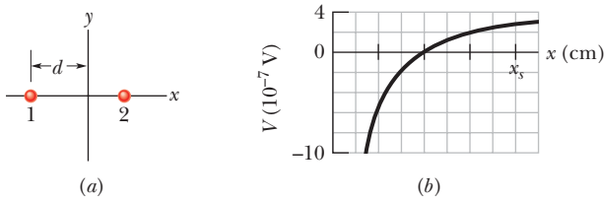



**Figure 24-39** Problem 18.

**••19** In Fig. 24-40, particles with the charges $q_1 = +5e$ and $q_2 = -15e$ are fixed in place with a separation of $d = 24.0$ cm. With electric potential defined to be $V = 0$ at infinity, what are the finite (a) positive and (b) negative values of $x$ at which the net electric potential on the $x$ axis is zero?

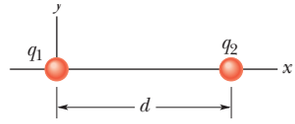



**Figure 24-40** Problems 19 and 20.

**••20** Two particles, of charges $q_1$ and $q_2$, are separated by distance $d$ in Fig. 24-40. The net electric field due to the particles is zero at $x = d/4$. With $V = 0$ at infinity, locate (in terms of $d$) any point on the $x$ axis (other than at infinity) at which the electric potential due to the two particles is zero.

## Module 24-4 Potential Due to an Electric Dipole

**•21** ILW The ammonia molecule $NH_3$ has a permanent electric dipole moment equal to 1.47 D, where 1 D = 1 debye unit = $3.34 \times 10^{-30}$ C·m. Calculate the electric potential due to an ammonia molecule at a point 52.0 nm away along the axis of the dipole. (Set $V = 0$ at infinity.)

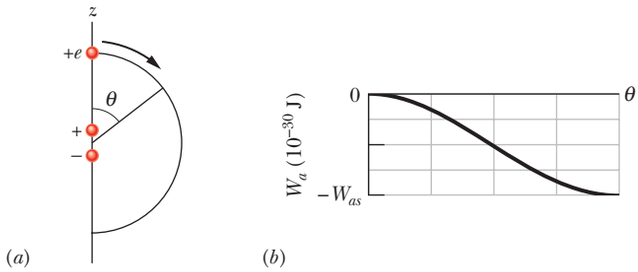



**Figure 24-41** Problem 22.

**••22** In Fig. 24-41*a*, a particle of elementary charge $+e$ is initially at coordinate $z = 20$ nm on the dipole axis (here a $z$ axis) through an electric dipole, on the positive side of the dipole. (The origin of $z$ is at the center of the dipole.) The particle is then moved along a circular path around the dipole center until it is at coordinate $z = -20$ nm, on the negative side of the dipole axis. Figure 24-41*b* gives the work $W_a$ done by the force moving the particle versus the angle $\theta$ that locates the particle relative to the positive direction of the $z$ axis. The scale of the vertical axis is set by $W_{as} = 4.0 \times 10^{-30}$ J. What is the magnitude of the dipole moment?

## Module 24-5 Potential Due to a Continuous Charge Distribution

**•23** (a) Figure 24-42*a* shows a nonconducting rod of length $L = 6.00$ cm and uniform linear charge density $\lambda = +3.68$ pC/m. Assume that the electric potential is defined to be $V = 0$ at infinity. What is $V$ at point $P$ at distance $d = 8.00$ cm along the rod's perpendicular bisector? (b) Figure 24-42*b* shows an identical rod except that one half is now negatively charged. Both halves have a linear charge density of magnitude 3.68 pC/m. With $V = 0$ at infinity, what is $V$ at $P$?

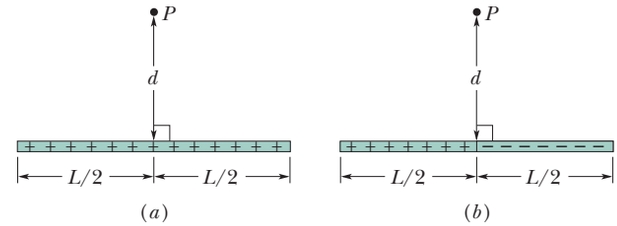



**Figure 24-42** Problem 23.

**•24** In Fig. 24-43, a plastic rod having a uniformly distributed charge $Q = -25.6$ pC has been bent into a circular arc of radius $R = 3.71$ cm and central angle $\phi = 120°$. With $V = 0$ at infinity, what is the electric potential at $P$, the center of curvature of the rod?

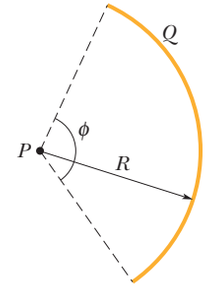



**Figure 24-43** Problem 24.

**•25** A plastic rod has been bent into a circle of radius $R = 8.20$ cm. It has a charge $Q_1 = +4.20$ pC uniformly distributed along one-quarter of its circumference and a charge $Q_2 = -6Q_1$ uniformly distributed along the rest of the circumference (Fig. 24-44). With $V = 0$ at infinity, what is the electric potential at (a) the center $C$ of the circle and (b) point $P$, on the central axis of the circle at distance $D = 6.71$ cm from the center?

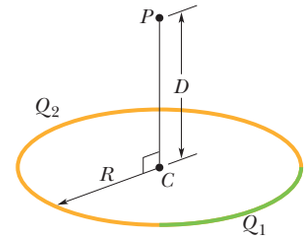



**Figure 24-44** Problem 25.

**••26** GO Figure 24-45 shows a thin rod with a uniform charge density of 2.00 $\mu$C/m. Evaluate the electric potential at point $P$ if $d = D = L/4.00$. Assume that the potential is zero at infinity.

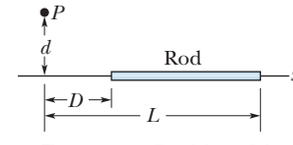



**Figure 24-45** Problem 26.

••27 In Fig. 24-46, three thin plastic rods form quarter-circles with a common center of curvature at the origin. The uniform charges on the three rods are $Q_1 = +30$ nC, $Q_2 = +3.0Q_1$, and $Q_3 = -8.0Q_1$. What is the net electric potential at the origin due to the rods?

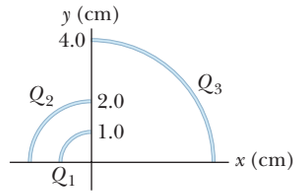



**Figure 24-46** Problem 27.

••28 GO Figure 24-47 shows a thin plastic rod of length $L = 12.0$ cm and uniform positive charge $Q = 56.1$ fC lying on an $x$ axis. With $V = 0$ at infinity, find the electric potential at point $P_1$ on the axis, at distance $d = 2.50$ cm from the rod.

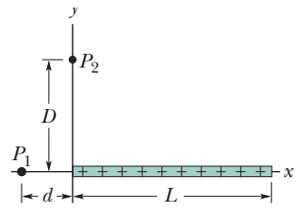



**Figure 24-47** Problems 28, 33, 38, and 40.

••29 In Fig. 24-48, what is the net electric potential at the origin due to the circular arc of charge $Q_1 = +7.21$ pC and the two particles of charges $Q_2 = 4.00Q_1$ and $Q_3 = -2.00Q_1$? The arc's center of curvature is at the origin and its radius is $R = 2.00$ m; the angle indicated is $\theta = 20.0°$.

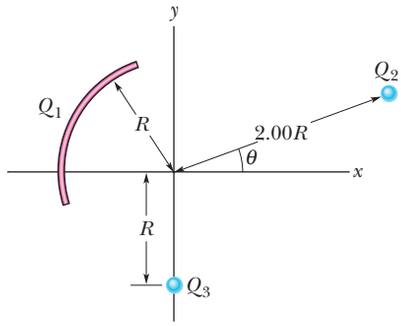



**Figure 24-48** Problem 29.

••30 GO The smiling face of Fig. 24-49 consists of three items:

1. a thin rod of charge $-3.0\ \mu$C that forms a full circle of radius 6.0 cm;
2. a second thin rod of charge $2.0\ \mu$C that forms a circular arc of radius 4.0 cm, subtending an angle of 90° about the center of the full circle;
3. an electric dipole with a dipole moment that is perpendicular to a radial line and has a magnitude of $1.28 \times 10^{-21}$ C·m.

What is the net electric potential at the center?

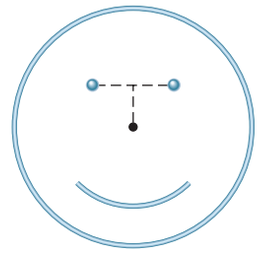


**Figure 24-49** Problem 30.

••31 SSM WWW A plastic disk of radius $R = 64.0$ cm is charged on one side with a uniform surface charge density $\sigma = 7.73$ fC/m$^2$, and then three quadrants of the disk are removed. The remaining quadrant is shown in Fig. 24-50. With $V = 0$ at infinity, what is the potential due to the remaining quadrant at point $P$, which is on the central axis of the original disk at distance $D = 25.9$ cm from the original center?

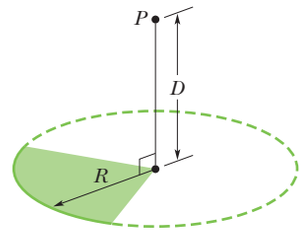



**Figure 24-50** Problem 31.

•••32 GO A nonuniform linear charge distribution given by $\lambda = bx$, where $b$ is a constant, is located along an $x$ axis from $x = 0$ to $x = 0.20$ m. If $b = 20$ nC/m$^2$ and $V = 0$ at infinity, what is the electric potential at (a) the origin and (b) the point $y = 0.15$ m on the $y$ axis?

•••33 GO The thin plastic rod shown in Fig. 24-47 has length $L = 12.0$ cm and a nonuniform linear charge density $\lambda = cx$, where $c = 28.9$ pC/m$^2$. With $V = 0$ at infinity, find the electric potential at point $P_1$ on the axis, at distance $d = 3.00$ cm from one end.

## Module 24-6 Calculating the Field from the Potential

•34 Two large parallel metal plates are 1.5 cm apart and have charges of equal magnitudes but opposite signs on their facing surfaces. Take the potential of the negative plate to be zero. If the potential halfway between the plates is then +5.0 V, what is the electric field in the region between the plates?

•35 The electric potential at points in an $xy$ plane is given by $V = (2.0\ \text{V/m}^2)x^2 - (3.0\ \text{V/m}^2)y^2$. In unit-vector notation, what is the electric field at the point (3.0 m, 2.0 m)?

•36 The electric potential $V$ in the space between two flat parallel plates 1 and 2 is given (in volts) by $V = 1500x^2$, where $x$ (in meters) is the perpendicular distance from plate 1. At $x = 1.3$ cm, (a) what is the magnitude of the electric field and (b) is the field directed toward or away from plate 1?

••37 SSM What is the magnitude of the electric field at the point $(3.00\hat{\mathbf{i}} - 2.00\hat{\mathbf{j}} + 4.00\hat{\mathbf{k}})$ m if the electric potential in the region is given by $V = 2.00xyz^2$, where $V$ is in volts and coordinates $x$, $y$, and $z$ are in meters?

••38 Figure 24-47 shows a thin plastic rod of length $L = 13.5$ cm and uniform charge 43.6 fC. (a) In terms of distance $d$, find an expression for the electric potential at point $P_1$. (b) Next, substitute variable $x$ for $d$ and find an expression for the magnitude of the component $E_x$ of the electric field at $P_1$. (c) What is the direction of $E_x$ relative to the positive direction of the $x$ axis? (d) What is the value of $E_x$ at $P_1$ for $x = d = 6.20$ cm? (e) From the symmetry in Fig. 24-47, determine $E_y$ at $P_1$.

••39 An electron is placed in an $xy$ plane where the electric potential depends on $x$ and $y$ as shown, for the coordinate axes, in Fig. 24-51 (the potential does not depend on $z$). The scale of the vertical axis is set by $V_s = 500$ V. In unit-vector notation, what is the electric force on the electron?

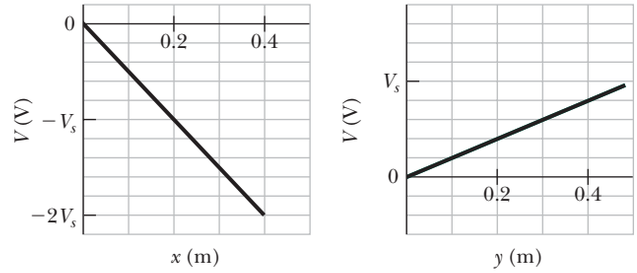



**Figure 24-51** Problem 39.

•••40 GO The thin plastic rod of length $L = 10.0$ cm in Fig. 24-47 has a nonuniform linear charge density $\lambda = cx$, where $c = 49.9$ pC/m$^2$. (a) With $V = 0$ at infinity, find the electric potential at point $P_2$ on the $y$ axis at $y = D = 3.56$ cm. (b) Find the electric field component $E_y$ at $P_2$. (c) Why cannot the field component $E_x$ at $P_2$ be found using the result of (a)?

### Module 24-7 Electric Potential Energy of a System of Charged Particles

•41 A particle of charge +7.5 $\mu$C is released from rest at the point $x = 60$ cm on an $x$ axis. The particle begins to move due to the presence of a charge $Q$ that remains fixed at the origin. What is the kinetic energy of the particle at the instant it has moved 40 cm if (a) $Q = +20$ $\mu$C and (b) $Q = -20$ $\mu$C?

•42 (a) What is the electric potential energy of two electrons separated by 2.00 nm? (b) If the separation increases, does the potential energy increase or decrease?

•43 SSM ILW WWW How much work is required to set up the arrangement of Fig. 24-52 if $q = 2.30$ pC, $a = 64.0$ cm, and the particles are initially infinitely far apart and at rest?

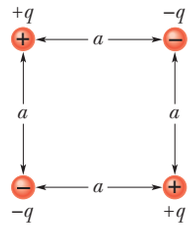


Figure 24-52 Problem 43.

•44 In Fig. 24-53, seven charged particles are fixed in place to form a square with an edge length of 4.0 cm. How much work must we do to bring a particle of charge +6$e$ initially at rest from an infinite distance to the center of the square?

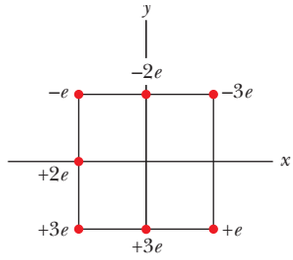


Figure 24-53 Problem 44.

••45 ILW A particle of charge $q$ is fixed at point $P$, and a second particle of mass $m$ and the same charge $q$ is initially held a distance $r_1$ from $P$. The second particle is then released. Determine its speed when it is a distance $r_2$ from $P$. Let $q = 3.1$ $\mu$C, $m = 20$ mg, $r_1 = 0.90$ mm, and $r_2 = 2.5$ mm.

••46 A charge of −9.0 nC is uniformly distributed around a thin plastic ring lying in a $yz$ plane with the ring center at the origin. A −6.0 pC particle is located on the $x$ axis at $x = 3.0$ m. For a ring radius of 1.5 m, how much work must an external force do on the particle to move it to the origin?

••47 GO What is the *escape speed* for an electron initially at rest on the surface of a sphere with a radius of 1.0 cm and a uniformly distributed charge of $1.6 \times 10^{-15}$ C? That is, what initial speed must the electron have in order to reach an infinite distance from the sphere and have zero kinetic energy when it gets there?

••48 A thin, spherical, conducting shell of radius $R$ is mounted on an isolating support and charged to a potential of −125 V. An electron is then fired directly toward the center of the shell, from point $P$ at distance $r$ from the center of the shell ($r \gg R$). What initial speed $v_0$ is needed for the electron to just reach the shell before reversing direction?

••49 GO Two electrons are fixed 2.0 cm apart. Another electron is shot from infinity and stops midway between the two. What is its initial speed?

••50 In Fig. 24-54, how much work must we do to bring a particle, of charge $Q = +16e$ and initially at rest, along the dashed line from infinity to the indicated point near two fixed particles of charges $q_1 = +4e$ and $q_2 = -q_1/2$? Distance $d = 1.40$ cm, $\theta_1 = 43°$, and $\theta_2 = 60°$.

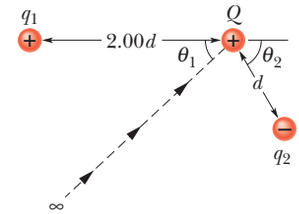


Figure 24-54 Problem 50.

••51 GO In the rectangle of Fig. 24-55, the sides have lengths 5.0 cm and 15 cm, $q_1 = -5.0$ $\mu$C, and $q_2 = +2.0$ $\mu$C. With $V = 0$ at infinity, what is the electric potential at (a) corner $A$ and (b) corner $B$? (c) How much work is required to move a charge $q_3 = +3.0$ $\mu$C from $B$ to $A$ along a diagonal of the rectangle? (d) Does this work increase or decrease the electric potential energy of the three-charge system? Is more, less, or the same work required if $q_3$ is moved along a path that is (e) inside the rectangle but not on a diagonal and (f) outside the rectangle?



Figure 24-55 Problem 51.

••52 Figure 24-56$a$ shows an electron moving along an electric dipole axis toward the negative side of the dipole. The dipole is fixed in place. The electron was initially very far from the dipole, with kinetic energy 100 eV. Figure 24-56$b$ gives the kinetic energy $K$ of the electron versus its distance $r$ from the dipole center. The scale of the horizontal axis is set by $r_s = 0.10$ m. What is the magnitude of the dipole moment?

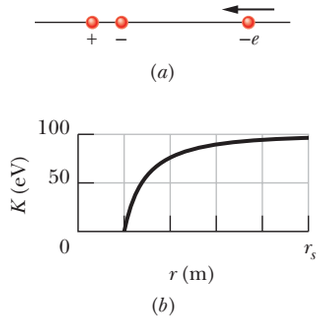


Figure 24-56 Problem 52.

••53 Two tiny metal spheres $A$ and $B$, mass $m_A = 5.00$ g and $m_B = 10.0$ g, have equal positive charge $q = 5.00$ $\mu$C. The spheres are connected by a massless nonconducting string of length $d = 1.00$ m, which is much greater than the radii of the spheres. (a) What is the electric potential energy of the system? (b) Suppose you cut the string. At that instant, what is the acceleration of each sphere? (c) A long time after you cut the string, what is the speed of each sphere?

••54 GO A positron (charge +$e$, mass equal to the electron mass) is moving at $1.0 \times 10^7$ m/s in the positive direction of an $x$ axis when, at $x = 0$, it encounters an electric field directed along the $x$ axis. The electric potential $V$ associated with the field is given in Fig. 24-57. The scale of the vertical axis is set by $V_s = 500.0$ V. (a) Does the positron emerge from the field at $x = 0$ (which means its motion is reversed) or at $x = 0.50$ m (which means its motion is not reversed)? (b) What is its speed when it emerges?

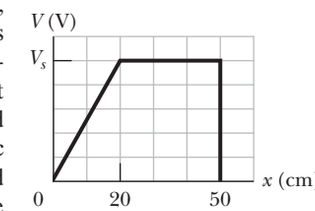


Figure 24-57 Problem 54.

••55 An electron is projected with an initial speed of $3.2 \times 10^5$ m/s directly toward a proton that is fixed in place. If the electron is initially a great distance from the proton, at what distance from the proton is the speed of the electron instantaneously equal to twice the initial value?

••56 Particle 1 (with a charge of +5.0 $\mu$C) and particle 2 (with a charge of +3.0 $\mu$C) are fixed in place with separation $d = 4.0$ cm

on the $x$ axis shown in Fig. 24-58*a*. Particle 3 can be moved along the $x$ axis to the right of particle 2. Figure 24-58*b* gives the electric potential energy $U$ of the three-particle system as a function of the $x$ coordinate of particle 3. The scale of the vertical axis is set by $U_s = 5.0$ J. What is the charge of particle 3?

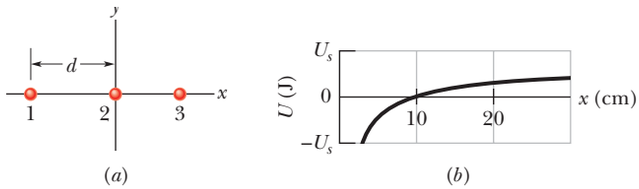



**Figure 24-58** Problem 56.

**••57** SSM Identical 50 $\mu$C charges are fixed on an $x$ axis at $x = \pm 3.0$ m. A particle of charge $q = -15$ $\mu$C is then released from rest at a point on the positive part of the $y$ axis. Due to the symmetry of the situation, the particle moves along the $y$ axis and has kinetic energy 1.2 J as it passes through the point $x = 0$, $y = 4.0$ m. (a) What is the kinetic energy of the particle as it passes through the origin? (b) At what negative value of $y$ will the particle momentarily stop?

**••58** GO *Proton in a well.* Figure 24-59 shows electric potential $V$ along an $x$ axis. The scale of the vertical axis is set by $V_s = 10.0$ V. A proton is to be released at $x = 3.5$ cm with initial kinetic energy 4.00 eV. (a) If it is initially moving in the negative direction of the axis, does it reach a turning point (if so, what is the $x$ coordinate of that point) or does it escape from the plotted region (if so, what is its speed at $x = 0$)? (b) If it is initially moving in the positive direction of the axis, does it reach a turning point (if so, what is the $x$ coordinate of that point) or does it escape from the plotted region (if so, what is its speed at $x = 6.0$ cm)? What are the (c) magnitude $F$ and (d) direction (positive or negative direction of the $x$ axis) of the electric force on the proton if the proton moves just to the left of $x = 3.0$ cm? What are (e) $F$ and (f) the direction if the proton moves just to the right of $x = 5.0$ cm?

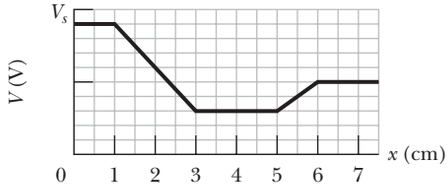



**Figure 24-59** Problem 58.

**••59** In Fig. 24-60, a charged particle (either an electron or a proton) is moving rightward between two parallel charged plates separated by distance $d = 2.00$ mm. The plate potentials are $V_1 = -70.0$ V and $V_2 = -50.0$ V. The particle is slowing from an initial speed of 90.0 km/s at the left plate. (a) Is the particle an electron or a proton? (b) What is its speed just as it reaches plate 2?

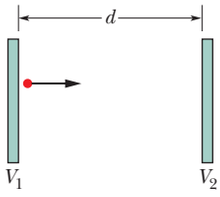



**Figure 24-60** Problem 59.

**••60** In Fig. 24-61*a*, we move an electron from an infinite distance to a point at distance $R = 8.00$ cm from a tiny charged ball. The move requires work $W = 2.16 \times 10^{-13}$ J by us. (a) What is the charge $Q$ on the ball? In Fig. 24-61*b*, the ball has been sliced up and the slices spread out so that an equal amount of charge is at the hour positions on a circular clock face of radius $R = 8.00$ cm. Now the electron is brought from an infinite distance to the center of the circle. (b) With that addition of the electron to the system of 12 charged particles, what is the change in the electric potential energy of the system?

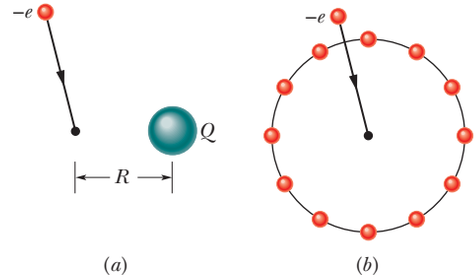



**Figure 24-61** Problem 60.

**•••61** Suppose $N$ electrons can be placed in either of two configurations. In configuration 1, they are all placed on the circumference of a narrow ring of radius $R$ and are uniformly distributed so that the distance between adjacent electrons is the same everywhere. In configuration 2, $N - 1$ electrons are uniformly distributed on the ring and one electron is placed in the center of the ring. (a) What is the smallest value of $N$ for which the second configuration is less energetic than the first? (b) For that value of $N$, consider any one circumference electron—call it $e_0$. How many other circumference electrons are closer to $e_0$ than the central electron is?

## Module 24-8 Potential of a Charged Isolated Conductor

**•62** Sphere 1 with radius $R_1$ has positive charge $q$. Sphere 2 with radius $2.00R_1$ is far from sphere 1 and initially uncharged. After the separated spheres are connected with a wire thin enough to retain only negligible charge, (a) is potential $V_1$ of sphere 1 greater than, less than, or equal to potential $V_2$ of sphere 2? What fraction of $q$ ends up on (b) sphere 1 and (c) sphere 2? (d) What is the ratio $\sigma_1/\sigma_2$ of the surface charge densities of the spheres?

**•63** SSM WWW Two metal spheres, each of radius 3.0 cm, have a center-to-center separation of 2.0 m. Sphere 1 has charge $+1.0 \times 10^{-8}$ C; sphere 2 has charge $-3.0 \times 10^{-8}$ C. Assume that the separation is large enough for us to say that the charge on each sphere is uniformly distributed (the spheres do not affect each other). With $V = 0$ at infinity, calculate (a) the potential at the point halfway between the centers and the potential on the surface of (b) sphere 1 and (c) sphere 2.

**•64** A hollow metal sphere has a potential of +400 V with respect to ground (defined to be at $V = 0$) and a charge of $5.0 \times 10^{-9}$ C. Find the electric potential at the center of the sphere.

**•65** SSM What is the excess charge on a conducting sphere of radius $r = 0.15$ m if the potential of the sphere is 1500 V and $V = 0$ at infinity?

**••66** Two isolated, concentric, conducting spherical shells have radii $R_1 = 0.500$ m and $R_2 = 1.00$ m, uniform charges $q_1 = +2.00$ $\mu$C and $q_2 = +1.00$ $\mu$C, and negligible thicknesses. What is the magnitude of the electric field $E$ at radial distance (a) $r = 4.00$ m, (b) $r = 0.700$ m, and (c) $r = 0.200$ m? With $V = 0$ at infinity, what is $V$ at (d) $r = 4.00$ m, (e) $r = 1.00$ m, (f) $r = 0.700$ m, (g) $r = 0.500$ m, (h) $r = 0.200$ m, and (i) $r = 0$? (j) Sketch $E(r)$ and $V(r)$.

**••67** A metal sphere of radius 15 cm has a net charge of $3.0 \times 10^{-8}$ C. (a) What is the electric field at the sphere's surface? (b) If $V = 0$ at infinity, what is the electric potential at the sphere's surface? (c) At what distance from the sphere's surface has the electric potential decreased by 500 V?

### Additional Problems

**68** Here are the charges and coordinates of two charged particles located in an $xy$ plane: $q_1 = +3.00 \times 10^{-6}$ C, $x = +3.50$ cm, $y = +0.500$ cm and $q_2 = -4.00 \times 10^{-6}$ C, $x = -2.00$ cm, $y = +1.50$ cm. How much work must be done to locate these charges at their given positions, starting from infinite separation?

**69** SSM A long, solid, conducting cylinder has a radius of 2.0 cm. The electric field at the surface of the cylinder is 160 N/C, directed radially outward. Let $A$, $B$, and $C$ be points that are 1.0 cm, 2.0 cm, and 5.0 cm, respectively, from the central axis of the cylinder. What are (a) the magnitude of the electric field at $C$ and the electric potential differences (b) $V_B - V_C$ and (c) $V_A - V_B$?

**70** *The chocolate crumb mystery.* This story begins with Problem 60 in Chapter 23. (a) From the answer to part (a) of that problem, find an expression for the electric potential as a function of the radial distance $r$ from the center of the pipe. (The electric potential is zero on the grounded pipe wall.) (b) For the typical volume charge density $\rho = -1.1 \times 10^{-3}$ C/m$^3$, what is the difference in the electric potential between the pipe's center and its inside wall? (The story continues with Problem 60 in Chapter 25.)

**71** SSM Starting from Eq. 24-30, derive an expression for the electric field due to a dipole at a point on the dipole axis.

**72** The magnitude $E$ of an electric field depends on the radial distance $r$ according to $E = A/r^4$, where $A$ is a constant with the unit volt–cubic meter. As a multiple of $A$, what is the magnitude of the electric potential difference between $r = 2.00$ m and $r = 3.00$ m?

**73** (a) If an isolated conducting sphere 10 cm in radius has a net charge of 4.0 $\mu$C and if $V = 0$ at infinity, what is the potential on the surface of the sphere? (b) Can this situation actually occur, given that the air around the sphere undergoes electrical breakdown when the field exceeds 3.0 MV/m?

**74** Three particles, charge $q_1 = +10\ \mu$C, $q_2 = -20\ \mu$C, and $q_3 = +30\ \mu$C, are positioned at the vertices of an isosceles triangle as shown in Fig. 24-62. If $a = 10$ cm and $b = 6.0$ cm, how much work must an external agent do to exchange the positions of (a) $q_1$ and $q_3$ and, instead, (b) $q_1$ and $q_2$?

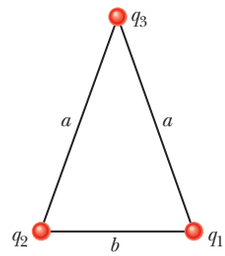


**Figure 24-62** Problem 74.

**75** An electric field of approximately 100 V/m is often observed near the surface of Earth. If this were the field over the entire surface, what would be the electric potential of a point on the surface? (Set $V = 0$ at infinity.)

**76** A Gaussian sphere of radius 4.00 cm is centered on a ball that has a radius of 1.00 cm and a uniform charge distribution. The total (net) electric flux through the surface of the Gaussian sphere is $+5.60 \times 10^4$ N·m$^2$/C. What is the electric potential 12.0 cm from the center of the ball?

**77** In a Millikan oil-drop experiment (Module 22-6), a uniform electric field of $1.92 \times 10^5$ N/C is maintained in the region between two plates separated by 1.50 cm. Find the potential difference between the plates.

**78** Figure 24-63 shows three circular, nonconducting arcs of radius $R = 8.50$ cm. The charges on the arcs are $q_1 = 4.52$ pC, $q_2 = -2.00q_1$, $q_3 = +3.00q_1$. With $V = 0$ at infinity, what is the net electric potential of the arcs at the common center of curvature?

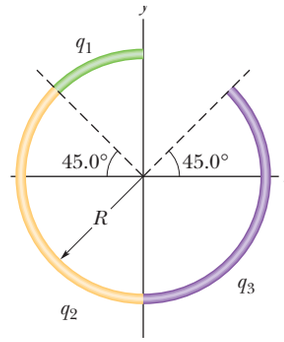


**Figure 24-63** Problem 78.

**79** An electron is released from rest on the axis of an electric dipole that has charge $e$ and charge separation $d = 20$ pm and that is fixed in place. The release point is on the positive side of the dipole, at distance $7.0d$ from the dipole center. What is the electron's speed when it reaches a point $5.0d$ from the dipole center?

**80** Figure 24-64 shows a ring of outer radius $R = 13.0$ cm, inner radius $r = 0.200R$, and uniform surface charge density $\sigma = 6.20$ pC/m$^2$. With $V = 0$ at infinity, find the electric potential at point $P$ on the central axis of the ring, at distance $z = 2.00R$ from the center of the ring.

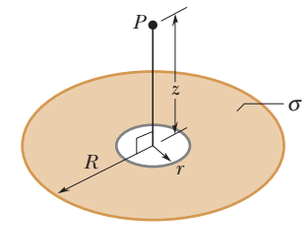


**Figure 24-64** Problem 80.

**81** GO *Electron in a well.* Figure 24-65 shows electric potential $V$ along an $x$ axis. The scale of the vertical axis is set by $V_s = 8.0$ V. An electron is to be released at $x = 4.5$ cm with initial kinetic energy 3.00 eV. (a) If it is initially moving in the negative direction of the axis, does it reach a turning point (if so, what is the $x$ coordinate of that point) or does it escape from the plotted region (if so, what is its speed at $x = 0$)? (b) If it is initially moving in the positive direction of the axis, does it reach a turning point (if so, what is the $x$ coordinate of that point) or does it escape from the plotted region (if so, what is its speed at $x = 7.0$ cm)? What are the (c) magnitude $F$ and (d) direction (positive or negative direction of the $x$ axis) of the electric force on the electron if the electron moves just to the left of $x = 4.0$ cm? What are (e) $F$ and (f) the direction if it moves just to the right of $x = 5.0$ cm?

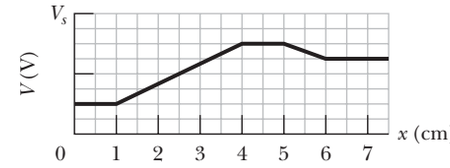


**Figure 24-65** Problem 81.

**82** (a) If Earth had a uniform surface charge density of 1.0 electron/m$^2$ (a very artificial assumption), what would its potential be? (Set $V = 0$ at infinity.) What would be the (b) magnitude and (c) direction (radially inward or outward) of the electric field due to Earth just outside its surface?

**83** In Fig. 24-66, point $P$ is at distance $d_1 = 4.00$ m from particle 1 ($q_1 = -2e$) and distance $d_2 = 2.00$ m from particle 2 ($q_2 = +2e$), with both particles fixed in place. (a) With $V = 0$ at infinity, what is $V$ at $P$? If we bring a particle of charge $q_3 = +2e$ from infinity to $P$, (b) how much work do we do and (c) what is the potential energy of the three-particle system?

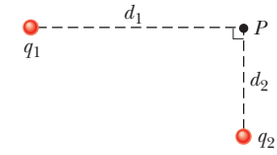


**Figure 24-66** Problem 83.

**84** A solid conducting sphere of radius 3.0 cm has a charge of 30 nC distributed uniformly over its surface. Let $A$ be a point 1.0 cm from the center of the sphere, $S$ be a point on the surface of the sphere, and $B$ be a point 5.0 cm from the center of the sphere. What are the electric potential differences (a) $V_S - V_B$ and (b) $V_A - V_B$?

**85** In Fig. 24-67, we move a particle of charge $+2e$ in from infinity to the $x$ axis. How much work do we do? Distance $D$ is 4.00 m.

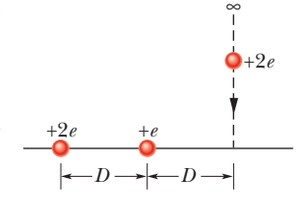


**Figure 24-67** Problem 85.

**86** Figure 24-68 shows a hemisphere with a charge of 4.00 $\mu$C distributed uniformly through its volume. The hemisphere lies on an $xy$ plane the way half a grapefruit might lie face down on a kitchen table. Point $P$ is located on the plane, along a radial line from the hemisphere's center of curvature, at radial distance 15 cm. What is the electric potential at point $P$ due to the hemisphere?

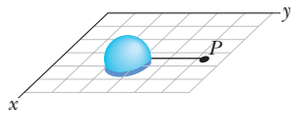


**Figure 24-68** Problem 86.

**87** SSM Three +0.12 C charges form an equilateral triangle 1.7 m on a side. Using energy supplied at the rate of 0.83 kW, how many days would be required to move one of the charges to the midpoint of the line joining the other two charges?

**88** Two charges $q = +2.0\ \mu$C are fixed a distance $d = 2.0$ cm apart (Fig. 24-69). (a) With $V = 0$ at infinity, what is the electric potential at point $C$? (b) You bring a third charge $q = +2.0\ \mu$C from infinity to $C$. How much work must you do? (c) What is the potential energy $U$ of the three-charge configuration when the third charge is in place?

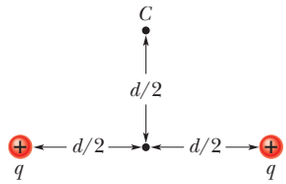


**Figure 24-69** Problem 88.

**89** Initially two electrons are fixed in place with a separation of 2.00 $\mu$m. How much work must we do to bring a third electron in from infinity to complete an equilateral triangle?

**90** A particle of positive charge $Q$ is fixed at point $P$. A second particle of mass $m$ and negative charge $-q$ moves at constant speed in a circle of radius $r_1$, centered at $P$. Derive an expression for the work $W$ that must be done by an external agent on the second particle to increase the radius of the circle of motion to $r_2$.

**91** Two charged, parallel, flat conducting surfaces are spaced $d = 1.00$ cm apart and produce a potential difference $\Delta V = 625$ V between them. An electron is projected from one surface directly toward the second. What is the initial speed of the electron if it stops just at the second surface?

**92** In Fig. 24-70, point $P$ is at the center of the rectangle. With $V = 0$ at infinity, $q_1 = 5.00$ fC, $q_2 = 2.00$ fC, $q_3 = 3.00$ fC, and $d = 2.54$ cm, what is the net electric potential at $P$ due to the six charged particles?

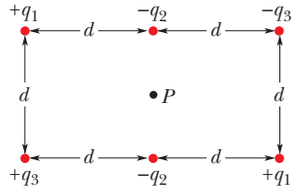


**Figure 24-70** Problem 92.

**93** SSM A uniform charge of +16.0 $\mu$C is on a thin circular ring lying in an $xy$ plane and centered on the origin. The ring's radius is 3.00 cm. If point $A$ is at the origin and point $B$ is on the $z$ axis at $z = 4.00$ cm, what is $V_B - V_A$?

**94** Consider a particle with charge $q = 1.50 \times 10^{-8}$ C, and take $V = 0$ at infinity. (a) What are the shape and dimensions of an equipotential surface having a potential of 30.0 V due to $q$ alone? (b) Are surfaces whose potentials differ by a constant amount (1.0 V, say) evenly spaced?

**95** SSM A thick spherical shell of charge $Q$ and uniform volume charge density $\rho$ is bounded by radii $r_1$ and $r_2 > r_1$. With $V = 0$ at infinity, find the electric potential $V$ as a function of distance $r$ from the center of the distribution, considering regions (a) $r > r_2$, (b) $r_2 > r > r_1$, and (c) $r < r_1$. (d) Do these solutions agree with each other at $r = r_2$ and $r = r_1$? (*Hint:* See Module 23-6.)

**96** A charge $q$ is distributed uniformly throughout a spherical volume of radius $R$. Let $V = 0$ at infinity. What are (a) $V$ at radial distance $r < R$ and (b) the potential difference between points at $r = R$ and the point at $r = 0$?

**97** SSM A solid copper sphere whose radius is 1.0 cm has a very thin surface coating of nickel. Some of the nickel atoms are radioactive, each atom emitting an electron as it decays. Half of these electrons enter the copper sphere, each depositing 100 keV of energy there. The other half of the electrons escape, each carrying away a charge $-e$. The nickel coating has an activity of $3.70 \times 10^8$ radioactive decays per second. The sphere is hung from a long, nonconducting string and isolated from its surroundings. (a) How long will it take for the potential of the sphere to increase by 1000 V? (b) How long will it take for the temperature of the sphere to increase by 5.0 K due to the energy deposited by the electrons? The heat capacity of the sphere is 14 J/K.

**98** In Fig. 24-71, a metal sphere with charge $q = 5.00\ \mu$C and radius $r = 3.00$ cm is concentric with a larger metal sphere with charge $Q = 15.0\ \mu$C and radius $R = 6.00$ cm. (a) What is the potential difference between the spheres? If we connect the spheres with a wire, what then is the charge on (b) the smaller sphere and (c) the larger sphere?

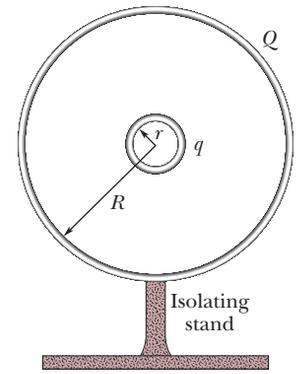


**Figure 24-71** Problem 98.

**99** (a) Using Eq. 24-32, show that the electric potential at a point on the central axis of a thin ring (of charge $q$ and radius $R$) and at distance $z$ from the ring is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{\sqrt{z^2 + R^2}}.$$

(b) From this result, derive an expression for the electric field magnitude $E$ at points on the ring's axis; compare your result with the calculation of $E$ in Module 22-4.

**100** An alpha particle (which has two protons) is sent directly toward a target nucleus containing 92 protons. The alpha particle has an initial kinetic energy of 0.48 pJ. What is the least center-to-center distance the alpha particle will be from the target nucleus, assuming the nucleus does not move?

**101** In the quark model of fundamental particles, a proton is composed of three quarks: two "up" quarks, each having charge $+2e/3$, and one "down" quark, having charge $-e/3$. Suppose that the three quarks are equidistant from one another. Take that separation distance to be $1.32 \times 10^{-15}$ m and calculate the electric potential energy of the system of (a) only the two up quarks and (b) all three quarks.

**102** A charge of $1.50 \times 10^{-8}$ C lies on an isolated metal sphere of radius 16.0 cm. With $V = 0$ at infinity, what is the electric potential at points on the sphere's surface?

**103** In Fig. 24-72, two particles of charges $q_1$ and $q_2$ are fixed to an $x$ axis. If a third particle, of charge $+6.0\ \mu$C, is brought from an infinite distance to point $P$, the three-particle system has the same electric potential energy as the original two-particle system. What is the charge ratio $q_1/q_2$?

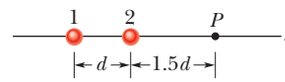


**Figure 24-72** Problem 103.